

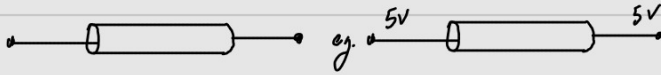
Circuit & Electronics → made of semiconductor

or Wattage

Power → 1 Watt

Electric Potential / Voltage → 1 Volt

whenever a charge moves
Current → 1 Ampere



$$I \propto V$$

$$\Delta V = IR$$

Just like Force $2N \rightarrow 5N$
(5-2)N = 3N →

Assuming positive elements power to be true

Passive Sign Convention

Current flows from +ve to -ve

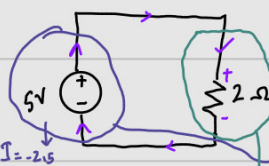
power → +5W → Consume
→ -5W → Supply

Conventional Current direction
is from high Voltage to low
voltage

Voltage can be +ve/-ve as charge can be +ve/-ve

Current +ve/-ve means opposite direction.

Power of Resistor is positive as it consumes power



$$P = \Delta V \cdot I$$

$$P = I^2 R \rightarrow \text{for resistor only}$$

$$\Delta V = IR$$

$$I = \frac{\Delta V}{R} = \frac{5}{2} = 2.5A$$

$$P = \Delta V \cdot I$$

$$= 5 \times 2.5 = 12.5W (\text{consuming})$$

Current should be from +ve to -ve from +ve to -ve (passive sign convention)

$$\therefore P = \Delta V I$$

$$= 5 \times 2.5$$

$$= 12.5W (\text{supplying})$$

$$F = -\frac{dV}{dx}, P = \frac{W}{t} = \frac{qV}{t} \quad P = VI, W = qV$$

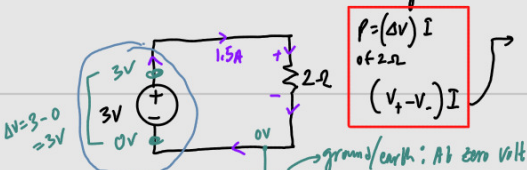
L-3

24/11/2024

Sunday

What do we mean when we want to find power?

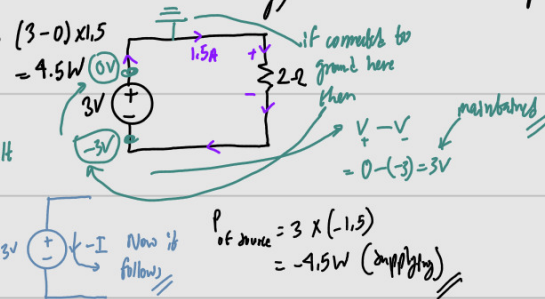
= It is not about the power of the whole circuit. Usually, we talk about the power of an appliance / component.



$$P = (\Delta V) I$$

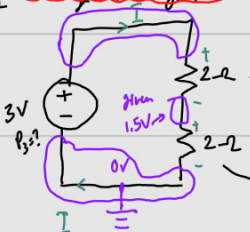
$$= (V_+ - V_-) I$$

ground/earth: At zero Volt



$$P_{\text{of source}} = 3 \times (-1.5) = -4.5W (\text{supplying})$$

eg Voltage Source given



$$I = \frac{3}{2+2} = 0.75$$

$$P_1 = (\Delta V) I$$

$$= (3-1.5) \times 0.75$$

$$= 1.125W$$

$$P_2 = (1.5-0) \times 0.75$$

$$= 1.125W$$

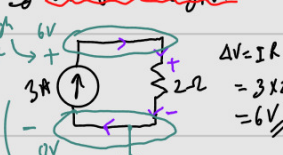
$$P_3 = 3 \times (-0.75)$$

$$= -2.25W$$

$$\text{OR } P_3 = -(P_1 + P_2)$$

$$= -2.25W$$

eg Current Source given



$$\Delta V = IR$$

$$= 3 \times 2$$

$$= 6V$$

$$P = (6-0) \times 3$$

$$= 18W$$

Source always will have -ve power

less powerful

more powerful

consumes power



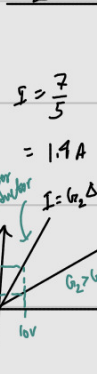
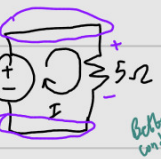
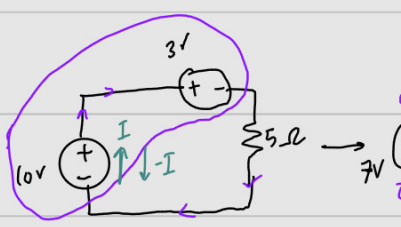
[From L-14]



What is the node voltage?

= no voltage, no current means no node voltage

L-4



$P_{10V} = (\Delta V)I$
 $= 10 \times -1.4$
 $= -14W$ (supplying)

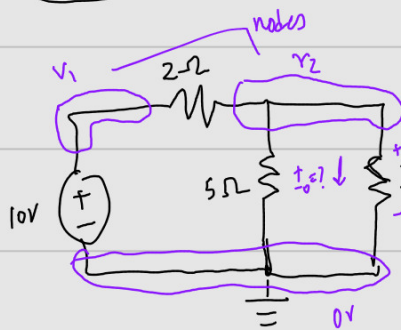
$P_{5\Omega} = (\Delta V)I$
 $= (V_+ - V_-)I$
 $= (7-0) \times 1.4$
 $= 9.8W$ (consuming)

$P_{3V} = 19 - 9.8$
 $= +9.2W$
 here voltage source is consuming power

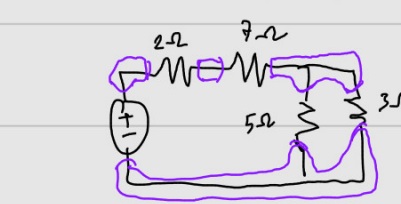
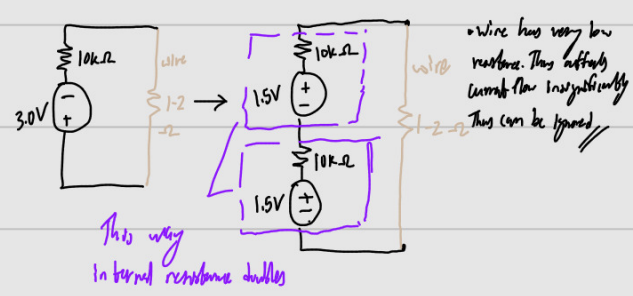
Ohm's story -> Ohm's I means intensity
 Whatstone

$I \propto \Delta V$
 $I = G_1 \Delta V$
 not $\Delta V \propto I$

$G_1 = \text{conductance}$
 how much current it allows
 $R = \frac{1}{G_1}$
 $I \propto \Delta V$
 $I = G_1 \Delta V$
 $I = \frac{1}{R} \Delta V$
 $I R = \Delta V$

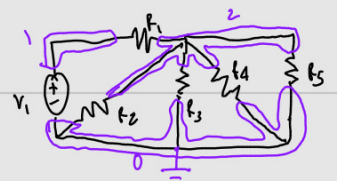


$I = \frac{\Delta V}{R} = \frac{10}{3}$ (wrong)
 # $I = \frac{\Delta V}{R} = \frac{V_2 - 0}{3} = \frac{V_2}{3} A$
 # For source, $\Delta V = 10 = V_1 - 0 = 10V$



How to identify nodes?
 = put opposite brackets on the terminal of every component. Then connect them without crossing a function

Parallel Connection

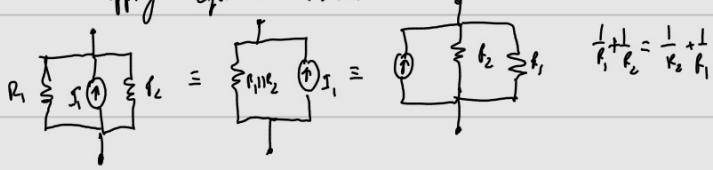


$R_4 || R_5?$
 $(2,0) \equiv (2,0)$
 same nodes
 yes

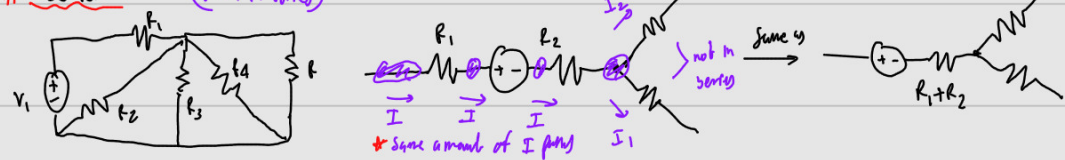
$R_1 || R_5?$
 $(1,2) \neq (2,0)$
 different nodes
 no

Series/Parallel Connection Advantages

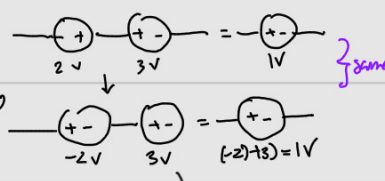
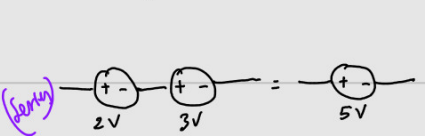
• Swapping • Equivalent Resistance



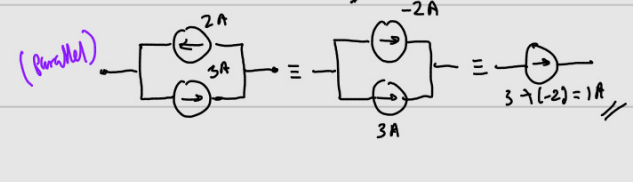
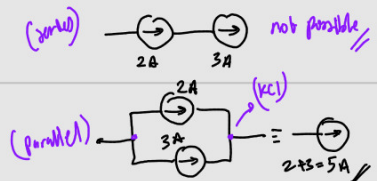
Series (Vanilla boxes)



Voltage Source



Current Source



In reality, the voltage sources have internal resistance which is very small. Thus a lot of current passes. Thus affecting the sources

Kirchoff's Current Law (KCL)

- Applied at a Node/Supernode
- Algebraic Sum
- Currents
- $\sum I = 0$

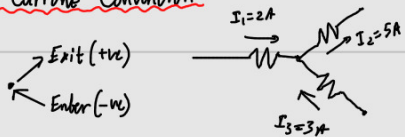
$$\sum I = \sum I_{\text{enter}} = \sum I_{\text{exit}}$$

$$0 = -\sum I_{\text{enter}} + \sum I_{\text{exit}}$$

$$\sum I_{\text{exit}} + \sum (-I_{\text{enter}}) = 0$$

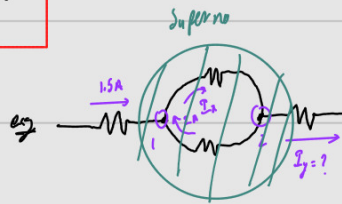
$$\sum I = 0$$

KCL Current Convention



$$I_1 + I_3 = I_2$$

$$-I_1 - I_3 + I_2 = 0$$



KCL @ Node 1

$$-1.5 - 2 + I_x = 0$$

$$\therefore I_x = 2 + 1.5 = 3.5A$$

KCL @ Node 2

$$-I_x + 2 + I_y = 0$$

$$-3.5 + 2 + I_y = 0$$

$$I_y = 1.5A$$

$$-1.5 + I_y = 0$$

$$I_y = 1.5A$$

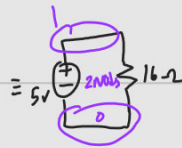
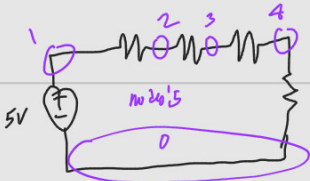
Actually what happens:

$$\text{Node 2! } -I_x + 2 + I_y = 0$$

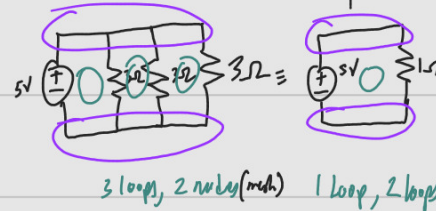
$$\text{Node 1! } -1.5 - 2 + I_x = 0$$

$$-1.5 + I_y = 0$$

$$I_y = 1.5A$$



$\therefore$ The node nodes decrease when resistors are in series



The no. of mesh decrease but nodes remain same in parallel connection of resistors

KCL

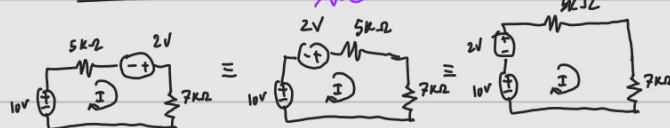
- (1) Node/Supernode
 - (2) Algebraic Sum
 - (3) Currents
 - (4) zero
- $$\sum I = 0$$

KVL

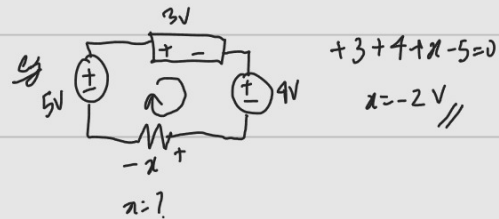
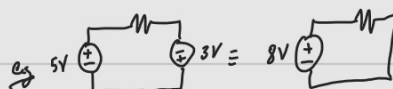
- (1) Mesh/Supernode
 - (2) Algebraic Sum
 - (3) Voltage
 - (4) zero
- $$\sum V = 0$$

$$L - 5.5$$

as before in series

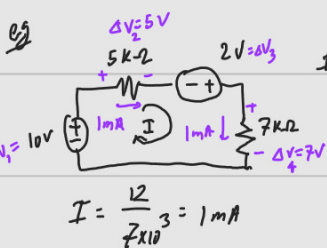


$$V, k\Omega, mA$$



$$+3 + 4 + x - 5 = 0$$

$$x = -2V$$



$$I = \frac{12}{7 \times 10^3} = 1mA$$

Mesh (Loop)

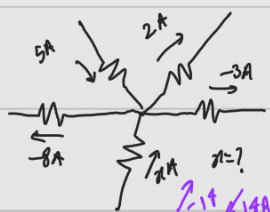
$$\sum (\Delta V) = 0$$

(Algebraic sum)

$$\# -\Delta V_1 + \Delta V_2 - \Delta V_3 + \Delta V_4 = 0$$

$$\Rightarrow -(10) + (5) - (2) + (7) = 0$$

$$\Rightarrow 0 = 0$$



KCL enter (-ve), exit (+ve)

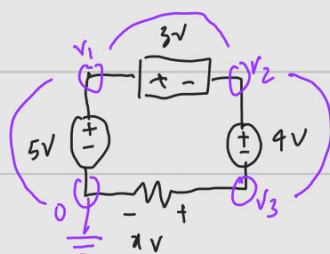
$$-x + (-3) + 2 - (5) + (-8) = 0$$

$$x = -14A$$

Voltage is of a node [1 node = 1 voltage]

voltage Difference is betⁿ two nodes [1 Element (Terminal) = 1 current]

Currents is of an element



$$V_1 - 0 = 5$$

$$V_1 - V_2 = 3$$

$$V_2 - V_3 = 4$$

$$V_3 - 0 = x$$

$$-4 - 3 + 5 - x = 0$$

$$V_3 - V_2 = -4$$

$$V_2 - V_1 = -3$$

$$V_1 - 0 = 5$$

$$0 - V_3 = -x$$

$$0 = -4 - 3 + 5 - x$$

Voltage Divider Rule (Only applicable for 2 resistors)

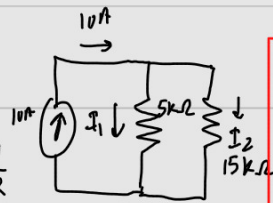
$R_1 = ?$ $R_2 = ?$

Series: 1) I same
2) V divide $\propto R$

$\Delta V_1 = 10V \times \frac{R_1}{R_1 + R_2}$
 $\Delta V_2 = 10V \times \frac{R_2}{R_1 + R_2}$

$\Delta V = V_1 - V_2$

Parallel: 1) V same
2) I divide $\propto \frac{1}{R}$

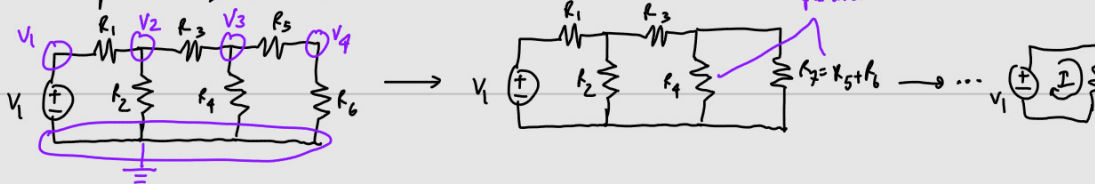


$$I_1 : I_2 = \frac{1}{R_1} : \frac{1}{R_2}$$

$$I_1 = 10 \times \frac{\frac{1}{R_1}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

$$I_2 = 10 \times \frac{\frac{1}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

Lab Exp 5: Series-Parallel



Brac University
Semester: Spring 2022
Course No: CSE250
Course Title: CIRCUITS AND ELECTRONICS
Section: 07 & 08
Faculty: SHS

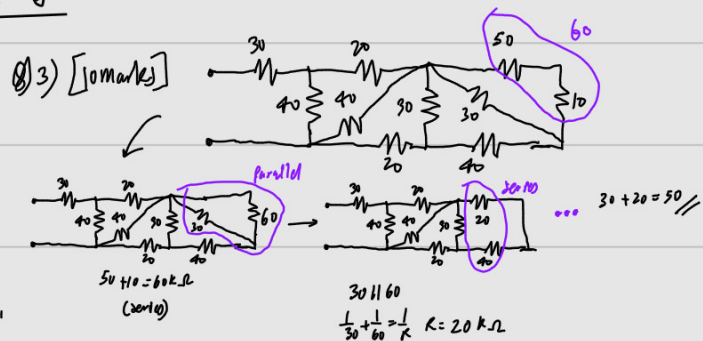
Quiz: 1
Full Marks: 20
Time: 20 minutes
Date: February 15, 2022

Question 1 [2 marks]

How many nodes are there in this circuit (including the ground node)? (2 marks)

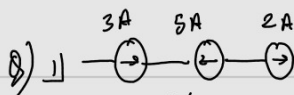
= 4 Nodes //

L-6

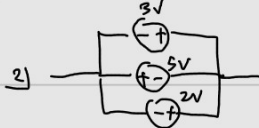


(i) $P = \Delta V I = V_+ - V_- = V_0 - V_1$
 $= -25 \times 10 \times 10^{-3} \text{ W}$

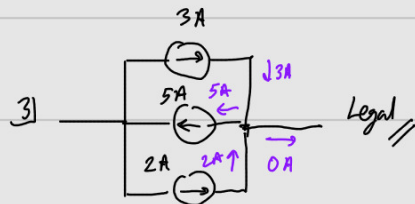
(ii) $P = \Delta V I = V_+ - V_- = V_0 - V_1$
 $= (-25) \times (-10 \times 10^{-3}) \text{ W}$



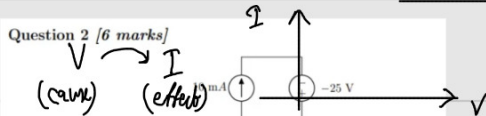
Illegal. Current sources can't be in series //



Illegal. Voltage source can't be in parallel //



I-V Characteristics



- What is the power of the current source (with appropriate $\pm$ sign)? (3 marks)
- What is the power of the voltage source (with appropriate $\pm$ sign)? (3 marks)

We've already seen this in Ohm's Law. $V = IR$

Ohm's Law

$$V = IR$$

$$I = \frac{1}{R} V + 0$$

$$y = mx + c$$

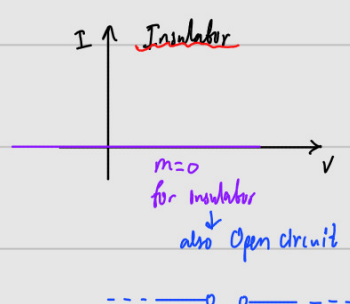


Resistor $\rightarrow$ I-V Characteristics

- 1) straight line
- 2) $m = \text{slope} = \frac{1}{R}$
- 3) $c = 0 \Rightarrow$ origin

$R = \infty \Omega$
(insulator)

$$m = \frac{1}{R} = \frac{1}{\lim_{R \rightarrow \infty}} = 0$$



Short Circuit

$R = 0 \Omega$
 $m = \frac{1}{R} = \frac{1}{\lim_{R \rightarrow 0}} = \infty$

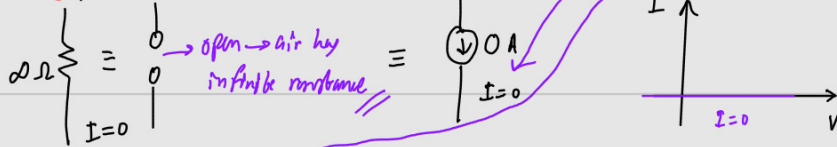
Characteristic eqⁿ of Ohm's Law: $V=IR$ [$I \propto V$]

only for resistor

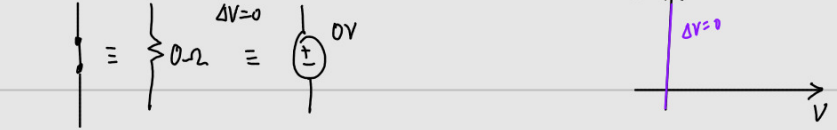
of voltage source

of current source

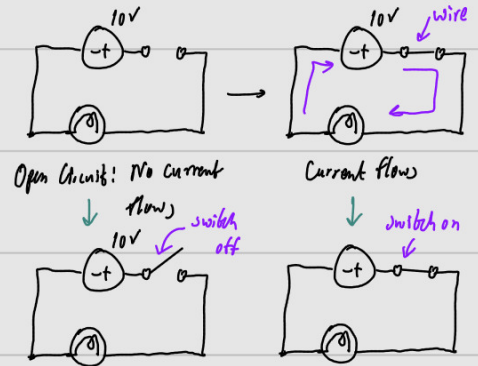
Open Circuit



Short Circuit

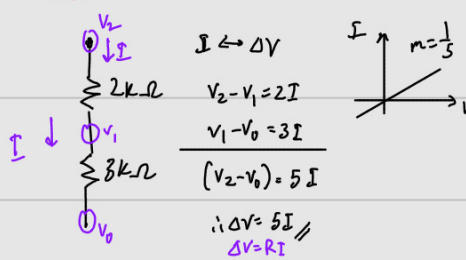


Equivalent Circuit has same characteristic eqⁿ

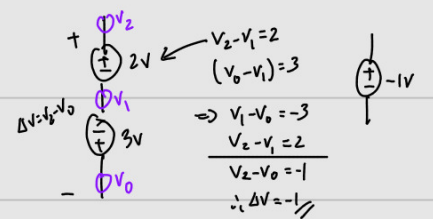
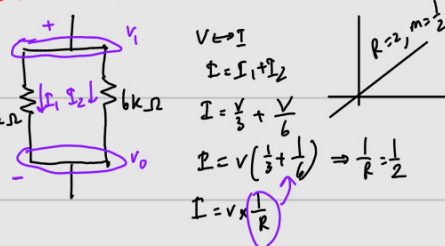


switch: switches betⁿ open & closed circuit. They're name "switch"

Series

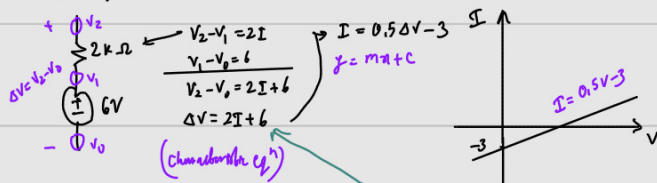


Parallel

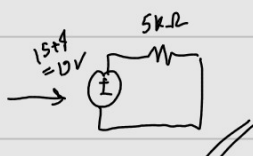
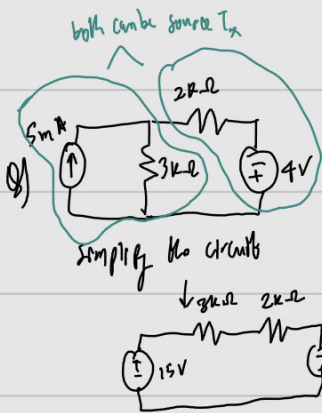
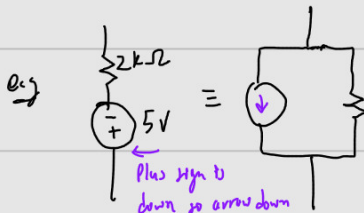
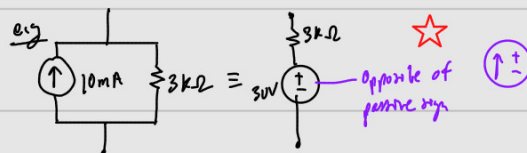
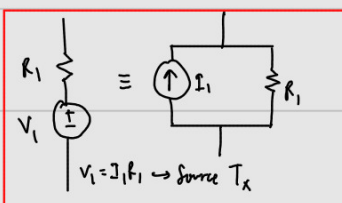
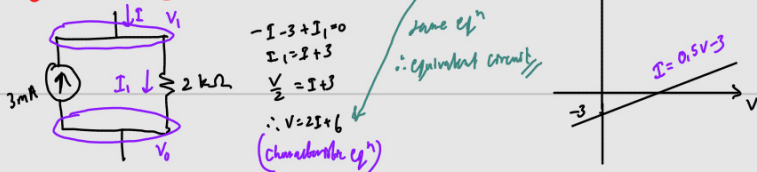


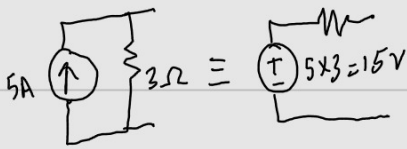
Source Transformation

Voltage Source & Resistor in Series

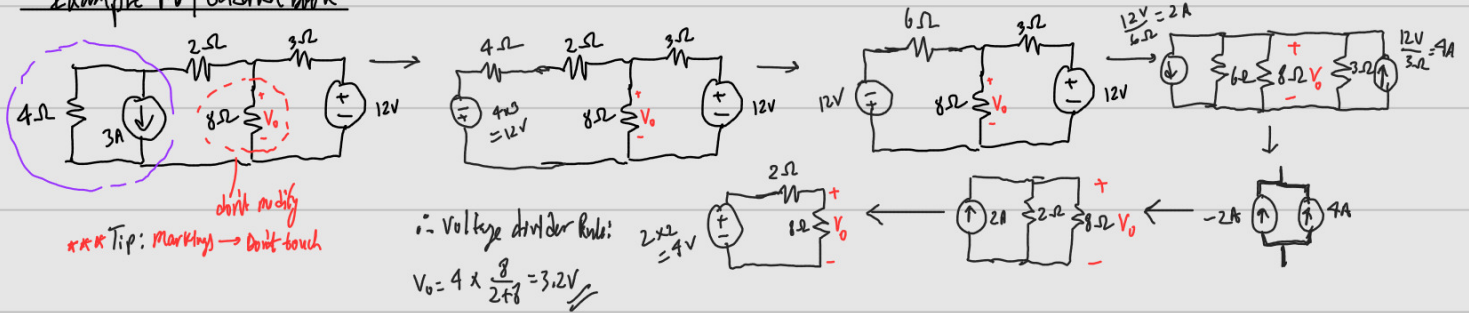


Current Source & Resistor in Parallel

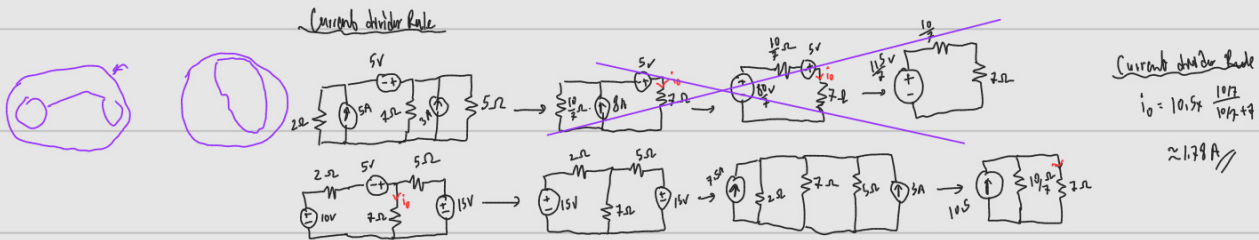




Example 4.6 | Superlump



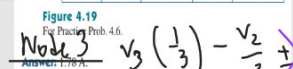
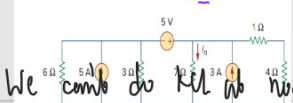
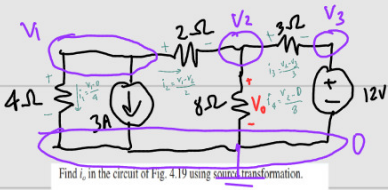
Practice Problem - 6



Drawback of Source Transformation

Work only when:

- Current source & resistor parallel
- Voltage source & resistor in series



extra variable
no need

Nodal Analysis (using KCL)

Step 1: Node Identification

(ground)

Step 2: Voltage source eqⁿ $4V = V_3 - 0 = 12$

Step 3: Resistance current (Ohm)

Step 4: Nodal

$$i_1 + i_2 + i_3 = 0$$

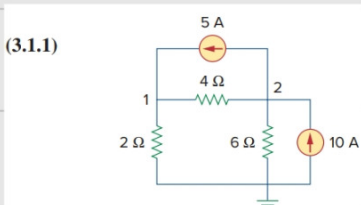
$$\Rightarrow \frac{V_1}{4} + \frac{V_1 - V_2}{2} + 3 = 0$$

$$\Rightarrow V_1 \left(\frac{1}{4} + \frac{1}{2} \right) - \frac{V_2}{2} + 3 = 0$$

Node 2: $V_2 \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{8} \right) - \frac{V_1}{2} - \frac{V_3}{3} - \frac{0}{8} = 0$

2 eq, 2 variables

Step 5: solve eqⁿ
(no need for RREF)



Step 1) Identify ref

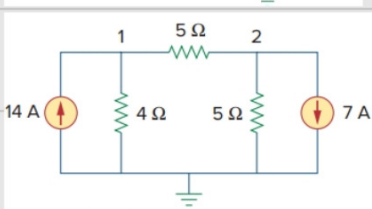
Step 2) Voltage source eqⁿ. None is none

Step 3) Nodal:

Step 4)

Node 1: $V_1 \left(\frac{1}{2} + \frac{1}{4} \right) - \frac{V_2}{2} - \frac{V_3}{4} - 5 = 0$

Node 2: $V_2 \left(\frac{1}{4} + \frac{1}{6} \right) - \frac{V_1}{2} - \frac{0}{6} - 10 + 5 = 0$



Node 1

$$V_1 \left(\frac{1}{5} + \frac{1}{4} \right) - \frac{V_2}{5} - \frac{0}{4} - 19 = 0$$

$$V_2 \left(\frac{1}{5} + \frac{1}{5} \right) - \frac{V_1}{5} - \frac{0}{5} + 7 = 0$$

$$V_1 = 30V, V_2 = -2.5V$$

Practice Problem 3.3

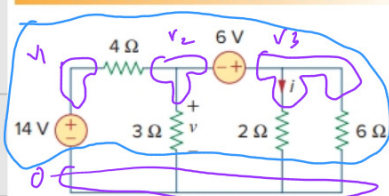


Figure 3.11
For Practice Prob. 3.3.

Find v and i .
Ans: -400 mV , 2.8 A

$$v_3 - v_2 = 6$$

$$v_1 - 0 = 14 \Rightarrow v_1 = 14$$

$$\left\{ v_1 \left(\frac{1}{4} \right) - \frac{v_2}{4} \right\} + \left\{ v_2 \left(\frac{1}{4} + \frac{1}{3} \right) - \frac{v_1}{4} \right\} + \left\{ v_3 \left(\frac{1}{2} + \frac{1}{6} \right) - \frac{v_2}{2} \right\} = 0$$

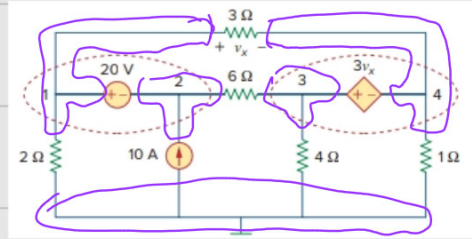
$$\frac{14}{4} + v_2 \left(-\frac{1}{4} + \frac{1}{3} + \frac{1}{3} \right) - \frac{14}{4} + v_3 \left(\frac{1}{2} + \frac{1}{6} \right) = 0$$

$$\Rightarrow \left\{ v_2 \left(\frac{1}{3} \right) - \frac{v_3}{4} \right\} + \left\{ v_3 \left(\frac{1}{2} + \frac{1}{6} \right) \right\} = 0 \Rightarrow v_2 \left(\frac{1}{3} + \frac{1}{2} + \frac{1}{6} \right) - \frac{14}{4} + v_3 \left(\frac{1}{2} + \frac{1}{6} \right) = 0$$

$$\Rightarrow v_2 \left(\frac{1}{3} \right) - \frac{14}{4} + (6 + v_2) \left(\frac{1}{2} + \frac{1}{6} \right) = 0 \Rightarrow v_2 = -0.4\text{ A} = -400\text{ mA}$$

$$v_3 = 6 + v_2 = 5.6\text{ V}$$

$$\# i = \frac{v_3}{2} = \frac{5.6}{2} = 2.8\text{ A}$$



Step 2: $v_1 - v_2 = 20$
 $v_3 - v_4 = 3v_x = 3(v_1 - v_4)$
 $v_4 = v_1 - v_4$

Supermesh at 1, 2: $v_1 \left(\frac{1}{2} + \frac{1}{3} \right) - \frac{0}{2} - \frac{v_2}{3} + v_2 \left(\frac{1}{6} \right) - \frac{v_3}{6} = 0$
 Supermesh at 3, 4: $v_3 \left(\frac{1}{4} + \frac{1}{6} \right) - \frac{0}{4} - \frac{v_2}{6} + v_4 \left(\frac{1}{4} + \frac{1}{3} \right) - \frac{0}{4} - \frac{v_1}{3} = 0$
 4 eqn, 4 variables $\rightarrow$ solve

L-10

Mesh Analysis

Nodal Analysis

- 1) $v_1, v_2, v_3, \dots$ (Node Voltage)
- 2) KCL eqn
- 3) Step 2: voltage source
- 4) Supernode [if there is floating voltage source]

Mesh Analysis

- 1) $i_1, i_2, i_3, \dots$ (mesh current)
- 2) KVL eqn
- 3) Step 2: Current Source
- 4) Supermesh [if there is/are common current source]

Take Mesh (Loop) direction clockwise always

Example 3.5

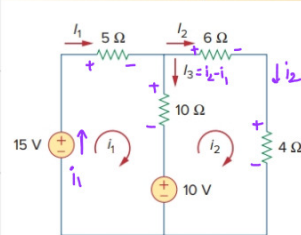


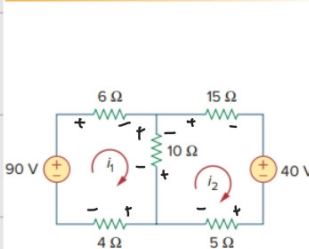
Figure 3.18
For Example 3.5.

Mesh 1: $-15 + 5i_1 + 10(i_1 - i_2) + 10 = 0$
 $\Rightarrow i_1(5 + 10) - 10i_2 - 15 + 10 = 0$
 Mesh 2: $-10 - 10(i_1 - i_2) + 6i_2 + 4i_2 = 0$
 $i_2(10 + 6 + 4) - 10i_1 - 10 = 0$
 $i_1 = \square, i_2 = \square$

Stepwise:

- Step 1: Mesh identification
 Step 2: Current source eqn
 Step 3/4: KVL at mesh
 Mesh 1: $i_1(5 + 10) - 10(i_2) - 15 + 10 = 0$
 Mesh 2: $i_2(6 + 4 + 10) - 10i_1 - 10 = 0$
 Step 5: solve

Practice Problem 3.5



Mesh 1: $i_1(6 + 10 + 4) - 6 \times 0 - 10 \times (-i_2) - 4 \times 0 - 20 = 0 \Rightarrow i_1(20) - 10i_2 = 20$
 Mesh 2: $i_2(15 + 5 + 10) - 15 \times 0 - 5 \times 0 - 10 \times (-i_1) + 40 = 0 \Rightarrow i_2(30) - 10i_1 = -40$
 $i_1 = 6.2\text{ A}, i_2 = -3.4\text{ A}$
 $i_1 = 4.6\text{ A}, i_2 = 0.2\text{ A}$

Example 3.6

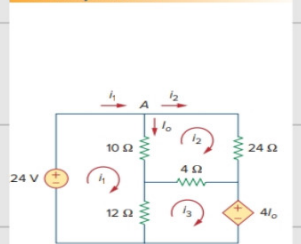


Figure 3.20
For Example 3.6.

- Step 1: Mesh identification
 Step 2: Current source eqn $i_3 = 4i_0$
 Step 3/4: Mesh 1: $i_1(10 + 12) - 10 \times i_2 - 12 \times i_3 - 24 = 0$
 Mesh 2: $i_2(24 + 4 + 10) - 24 \times 0 - 4 \times i_3 - 10 \times i_1 = 0$
 Mesh 3: $i_3(4 + 12) - 4 \times i_2 - 12 \times i_1 + 4i_0 = 0$
 3 eqn, 3 variables

Practice Problem 3.6

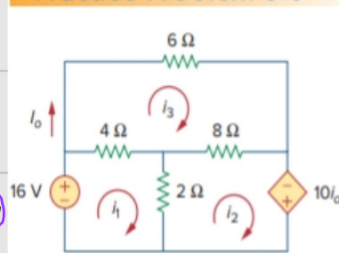
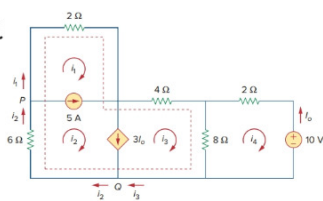


Figure 3.21
For Practice Prob. 3.6.

$i_0 = i_3$
 $i_1(4 + 6) - 4(i_2) - 2(i_3) - 16 = 0$
 $i_2(2 + 8) - 2(i_1) - 8(i_3) - 10i_0 = 0$
 $i_3(4 + 8) - 4(i_1) - 8(i_2) = 0$
 $i_1 = -2.57, i_2 = -7.71$
 $i_3 = -4\text{ A} = i_0$

For the circuit in Fig. 3.24, find i_1 to i_4 using mesh analysis.

Example 3.7



Step 2: $i_2 - i_1 = 5$

Step 3: $i_4 = 1$

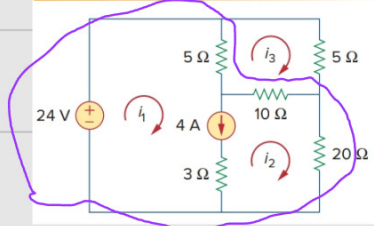
$i_2 - i_3 = 3I_0$

Mesh 4: $i_4(2+8) - 8(i_3) + 10 = 0$

$i_0 = -i_4$

Supermesh 1,2,3: $\{i_1(2) - 0\} + \{i_2(6)\} + \{i_3(4+8) - 4 \times 5 - 8 \times i_4\} = 0$

Practice Problem 3.7



Ans: $i_1 = 12.379$, $i_2 = 3749 \text{ mA}$, $i_3 = 3.284 \text{ A}$

Wrong Ans in book

$i_1 - i_2 = 4 \Rightarrow i_1 = 4 + i_2$

Mesh 3: $i_3(5+10+5) - 5 \times i_1 - 5 \times i_0 - 10 \times i_2 = 0$

$\Rightarrow i_3(20) - 5(4+i_2) - 10i_2 = 0 \Rightarrow -15i_2 + i_3(20) = 20$ (i)

Supermesh 1,2: $\{i_1(5+3) - 5 \times i_3 - 3 \times i_2 - 24\} + \{i_2(10+20+3) - 10 \times i_3 - 20(0) - 3(i_1)\} = 0$

$i_1(5) + i_2(30) + i_3(-15) = 24$

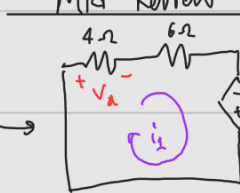
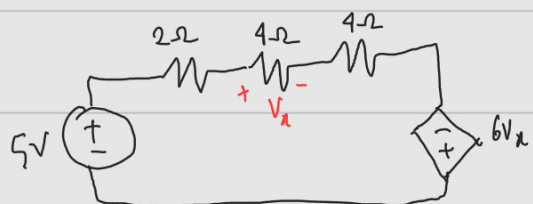
$5(4+i_2) + 30i_2 - 15i_3 = 24 \Rightarrow 35i_2 - 15i_3 = 4$ (ii)

$i_1 = 4.18$, $i_2 = 0.18$, $i_3 = 1.6$

26/11/2024

Tuesday

Mid Review



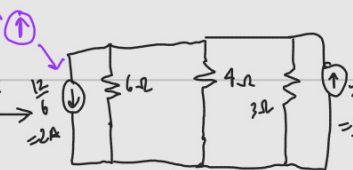
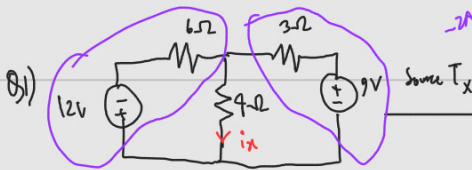
$V_A = 4i_1$ # $i_1(4+6) - (5+6V_A) = 0$

$\Rightarrow i_1 = \frac{V_A}{4}$

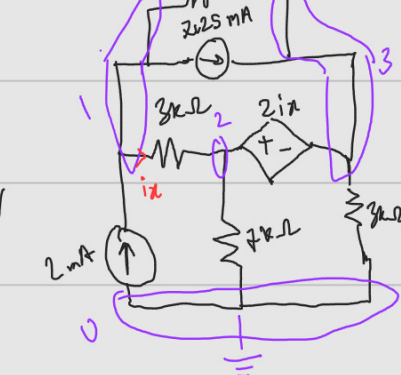
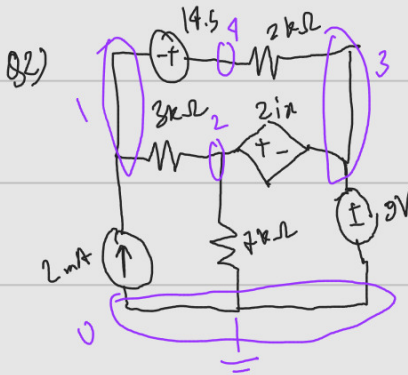
$\Rightarrow \frac{V_A}{4}(10) - (5+6V_A) = 0$

$\Rightarrow -3.5V_A - 5 = 0$

$\therefore V_A = \frac{5}{-3.5} = -\frac{10}{7} \approx -1.43 \text{ V}$



(Current divider rule)
$i_A = 1 \text{ K} \frac{2}{2+4} = \frac{1}{3} \approx 0.33 \text{ A}$
OR $(-i_A) \times 4 = i_A \times 2 \Rightarrow i_A = \frac{1}{3} \text{ A}$
 $1 - i_A = 2i_A$
 $1 = 3i_A$

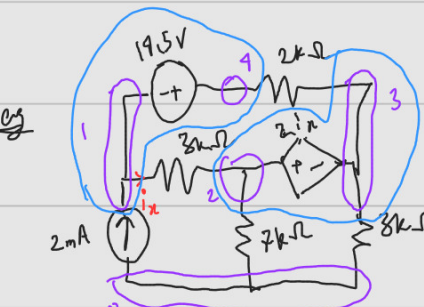


$i_A = \frac{V_1 - V_2}{3}$ (mA)

$V_2 - V_3 = 2i_A$

$V_1 \left(\frac{1}{3} + \frac{1}{2} \right) - \frac{V_2}{3} - \frac{V_3}{2} + 7.25 - 2 = 0$

$\left\{ V_2 \left(\frac{1}{3} + \frac{1}{7} \right) - \frac{V_1}{3} - \frac{0}{7} \right\} + \left\{ V_3 \left(\frac{1}{2} + \frac{1}{3} \right) - \frac{V_1}{2} - \frac{0}{3} - 7.25 \right\} = 0$



Nodal Analysis

$i_A = \frac{V_1 - V_2}{3}$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

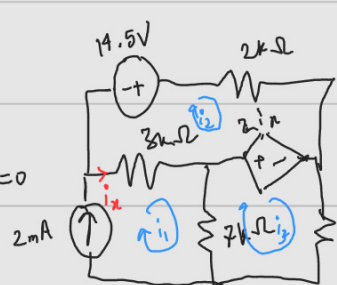
$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$

$2i_A = V_2 - V_3$



Mesh Analysis

$i_A = i_1 - i_2$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

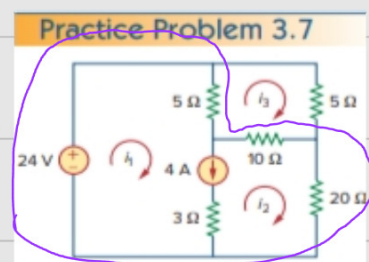
$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$

$i_1 = 2 \text{ mA}$



$i_1 - i_2 = 4 \text{ A}$

$i_3(5+5+10) - 5 \times i_1 - 5 \times i_0 - 10 \times i_2 = 0$

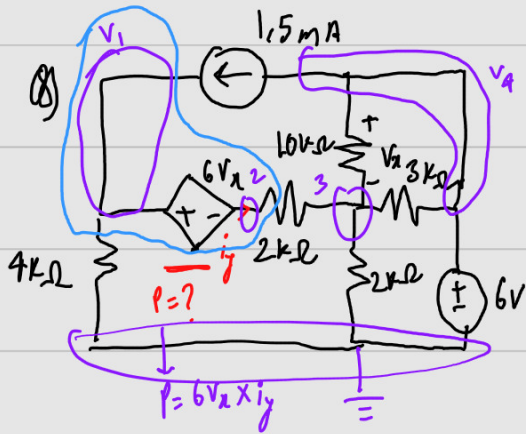
$\{i_1(5+3) - 5 \times i_3 - 3 \times i_2 - 24\} +$

$\{i_2(3+10+20) - 3 \times i_1 - 10 \times i_3\} = 0$

$\{i_2(3+10+20) - 3 \times i_1 - 10 \times i_3\} = 0$

$\{i_2(3+10+20) - 3 \times i_1 - 10 \times i_3\} = 0$

$\{i_2(3+10+20) - 3 \times i_1 - 10 \times i_3\} = 0$



Using Nodal Analysis

$$V_4 - 0 = 6 \Rightarrow V_4 = 6$$

$$V_4 - V_3 = V_x \Rightarrow V_3 = 6 - V_x$$

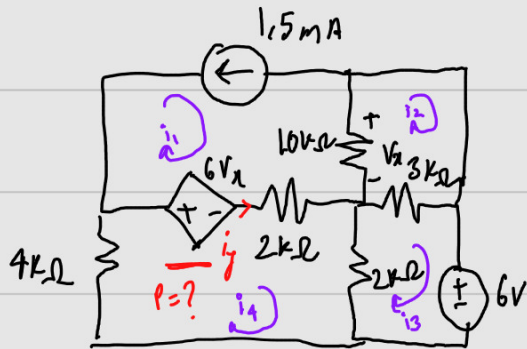
$$V_3 \left(\frac{1}{3} + \frac{1}{10} + \frac{1}{2} + \frac{1}{2} \right) - \frac{V_4}{3} - \frac{V_x}{10} - \frac{V_2}{2} - \frac{0}{2} = 0$$

$$\left\{ V_1 \left(\frac{1}{4} \right) - 1.5 \right\} + \left\{ V_2 \left(\frac{1}{2} \right) - \frac{V_3}{2} \right\} = 0$$

$$V_2 - V_3 = i_y \times 2$$

$$i_y = \boxed{} \text{ mA}$$

$$P = 6V_x \times i_y = \boxed{}$$



$$-i_1 = 1.5 \text{ mA}$$

$$i_2(3+10) - 3i_3 - 10i_1 = 0$$

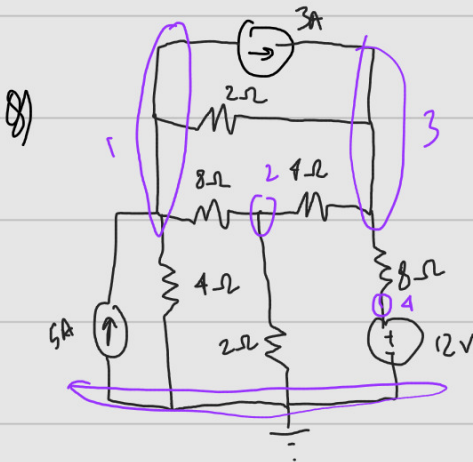
$$P = (6V_x) i_y //$$

$$i_y = i_4 - i_1$$

$$i_3(2+3) - 2(i_4) - 3(i_2) + 6 = 0$$

$$V_x = (i_2 - i_1) \times 10 \text{ V}$$

$$i_4(4+2+2) - 2(i_3) - 2(i_1) + 6V_x = 0$$



$$V_4 = 12 \text{ V}$$

$$\text{Node 1: } V_1 \left(\frac{1}{2} + \frac{1}{8} + \frac{1}{4} \right) - \frac{V_3}{2} - \frac{V_2}{8} - \frac{0}{4} - 5 + 3 = 0$$

$$V_1 = 10, V_2 = 9.03$$

$$\text{Node 2: } V_2 \left(\frac{1}{8} + \frac{1}{4} + \frac{1}{2} \right) - \frac{V_1}{8} - \frac{V_3}{4} - \frac{0}{2} = 0$$

$$V_3 = 12.27$$

$$\text{Node 3: } V_3 \left(\frac{1}{2} + \frac{1}{4} + \frac{1}{8} \right) - \frac{V_1}{2} - \frac{V_2}{8} - \frac{V_4}{4} - 3 = 0$$

$$\downarrow$$

$$-\frac{12}{8}$$

Problem-3

L-1

Problem 1

- How many electrons are there in 1C of charge?
- How much charge is represented by this number of electrons?
(a) 1.24×10^{18}
(b) 2.46×10^{19}
- What is the voltage between two points if 1.2 J of energy are required to move 20 μC between the two points?

Ans: (i) 6.25×10^{18} electrons
(ii) (a) -1.986 mC , (b) -3.941 C
(iii) 60 V

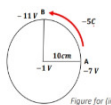
4) (i) $1e^- \rightarrow -1.602 \times 10^{-19} \text{ C}$
(ii) (a) $1e^- \rightarrow -1.6 \times 10^{-19} \text{ C}$
 $\therefore 1 \text{ C} \rightarrow \frac{1}{1.602 \times 10^{-19}} = 6.25 \times 10^{18} \text{ electrons}$
 $1.24 \times 10^{18} \rightarrow (1.24 \times 10^{18} \times -1.602 \times 10^{-19}) = -0.198648 \text{ C} \approx -198.65 \text{ mC}$

(b) $2.46 \times 10^{19} \rightarrow 2.46 \times 10^{19} \times (-1.602 \times 10^{-19}) = -3.94092 \text{ C} \approx -3.941 \text{ C}$

(iii) $V = \frac{W}{Q} = \frac{1.2}{20 \times 10^{-6}} = 60,000 \text{ V} = 60 \text{ kV}$

Problem 2

- At point A, there's a voltage of $V_A = 22 \text{ V}$. If a charge with $q = -2 \text{ C}$ moves from point A to another point B while moving it does a work of $W = 10 \text{ J}$, what's the voltage of point B?
- How much work must be done to transport the -5 C charge from point A to point B around the circle depicted in the diagram?



Ans: (i) $V_B = 27 \text{ V}$
(ii) $W = 30 \text{ J}$

2) (i) $W = q(V_B - V_A)$
 $-10 = -2(V_B - 22)$
 $\therefore V_B = 27 \text{ V}$
Work done by the charge is negative
(ii) $V_B = (-1) - (-1) = -12$
 $V_A = (-7) - (-1) = -8$
 $V_B - V_A = -12 - (-8) = -4$
 $W = -4 \times (-5) = 20 \text{ J}$

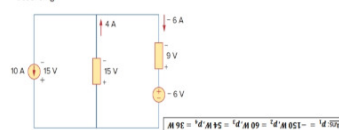
Problem 3

- Calculate the work that must be done in moving a -6 mC charge from point A to point B, where the potentials of points A and B are 20 V and -30 V respectively. Is the work done by the system?

3) $V_B - V_A = \frac{W}{q}$
 $-30 - (20) = \frac{W}{-6 \times 10^{-3}}$
 $\therefore W = 0.3 \text{ J}$

Problem 4

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

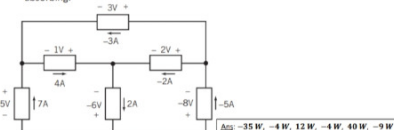


Ans: $P_1 = 15 \text{ W}$, $P_2 = 60 \text{ W}$, $P_3 = 54 \text{ W}$, $P_4 = 36 \text{ W}$

4) $P_1 = VI = 15 \times (-10) = -150 \text{ W}$ (Supplying)
 $P_2 = 15 \times 4 = 60 \text{ W}$
 $P_3 = 9 \times (-6) = 54 \text{ W}$ (Absorbing)
 $P_4 = -6 \times (-6) = 36 \text{ W}$

Problem 5

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

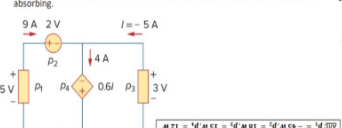


Ans: -35 W , -4 W , 12 W , -4 W , 40 W , -9 W

5) $P_1 = 5 \times (-7) = -35 \text{ W}$ (Supplying)
 $P_2 = 2 \times (-2) = -4 \text{ W}$ (Supplying)
 $P_3 = 3 \times (-3) = -9 \text{ W}$ (Supplying)
 $P_4 = 1 \times (-4) = -4 \text{ W}$ (Supplying)
 $P_5 = (-6) \times (-2) = 12 \text{ W}$ (Absorbing)
 $P_6 = -8 \times (-5) = 40 \text{ W}$ (Absorbing)

Problem 6

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

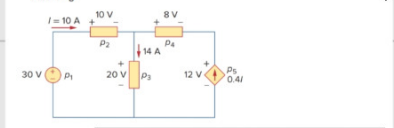


Ans: $P_1 = 15 \text{ W}$, $P_2 = 18 \text{ W}$, $P_3 = 15 \text{ W}$, $P_4 = 12 \text{ W}$

6) $P_1 = 5 \times (-9) = -45 \text{ W}$ (Supplying)
 $P_2 = 2 \times 9 = 18 \text{ W}$
 $P_3 = 3 \times (-5) = 15 \text{ W}$
 $P_4 = 0.6 \times (-4) = -2.4 \text{ W}$ (Supplying)

Problem 7

- Find power of each element in the network. Specify for each if it is supplying or absorbing.

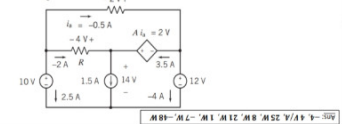


Ans: $P_1 = 30 \text{ W}$, $P_2 = 10 \text{ W}$, $P_3 = 280 \text{ W}$, $P_4 = -32 \text{ W}$, $P_5 = -9 \text{ W}$

7) $P_1 = 30 \times (-10) = -300 \text{ W}$
 $P_2 = 10 \times (10) = 100 \text{ W}$
 $P_3 = 20 \times (14) = 280 \text{ W}$
 $P_4 = 8 \times (-0.4 \text{ A}) = 8 \times (-0.4 \times 10) = -32 \text{ W}$
 $P_5 = 12 \times (-0.4 \text{ A}) = 12 \times (-0.4 \times 10) = -9 \text{ W}$

Problem 8

- Determine the value of the resistance R and the gain A of the CCVS.
- Find power of each element in the network. Specify for each if it is supplying or absorbing.



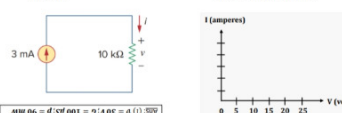
Ans: $R = 2 \Omega$, $A = -4$, $P_1 = 14 \text{ W}$, $P_2 = 48 \text{ W}$, $P_3 = 25 \text{ W}$

(a) $R = \frac{4}{2} = 2 \Omega$
 $A i_a = 2$
 $A = \frac{2}{-0.5} = -4$
(b) $P_{2V} = (2) \times (-0.5) = -1 \text{ W}$
 $P_R = 4 \times (-2) = 8 \text{ W}$
 $P_{10V} = 10 \times 2.5 = 25 \text{ W}$
 $P_{1V} = 1 \times 1.5 = 1.5 \text{ W}$
 $P_{12V} = 12 \times (-4) = -48 \text{ W}$

L-2

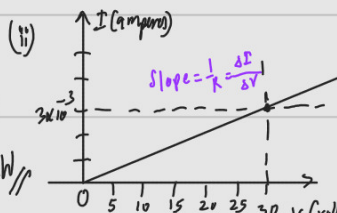
Problem 1

- For the circuit shown below, calculate the voltage v , the conductance G , and the power p .
- Draw the $I-v$ characteristics of a $10 \text{ k}\Omega$ resistor using the following template. Label the axes appropriately.



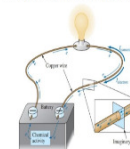
Ans: $v = 30 \text{ V}$, $G = 100 \text{ mS}$, $p = 90 \text{ mW}$

1) (i) $V = 3 \times 10^3 \times 10 \times 10^{-3} = 30 \text{ V}$
 $G = \frac{1}{R} = \frac{1}{10 \times 10^3} = 1 \times 10^{-4} \text{ S} = 100 \text{ mS}$
 $P = VI = 30 \times 3 \times 10^{-3} = 0.09 \text{ W} = 90 \text{ mW}$



Problem 2

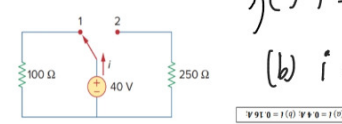
- Find the hot resistance of a light bulb rated 60 W , 120 V .



2) $P = VI$
 $I = \frac{60}{120} = 0.5 \text{ A}$
 $R = \frac{V}{I} = \frac{120}{0.5} = 240 \Omega$

Problem 3

- Calculate current i when the switch is in position 1.
- Find the current when the switch is in position 2.

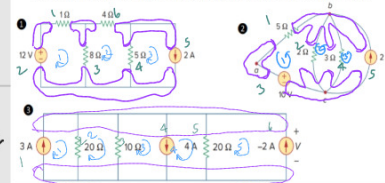


Ans: $i = 0.14 \text{ A}$, $i = 0.16 \text{ A}$

3) (a) $i = \frac{40}{250} = 0.16 \text{ A}$
(b) $i = \frac{40}{250} = 0.16 \text{ A}$

Problem-5

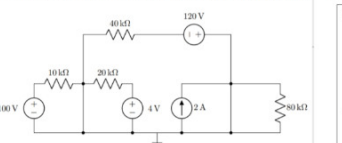
- Determine the number of branches, nodes, meshes, and loops in the following circuits.



1) Branches = 6, Nodes = 4, Meshes = 3, Loops = 6
2) b = 5, n = 3, m = 3, l = 6
3) b = 6, n = 2, m = 5, l = 15

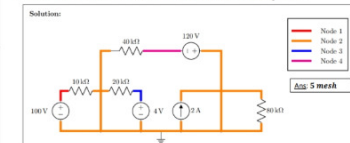
Problem 6

- Determine the number of nodes and meshes in the following circuit.



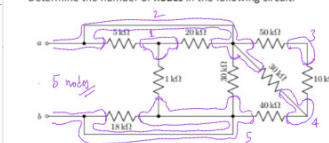
Problem 6: Solution

- Question: Determine the number of nodes in the following circuit.



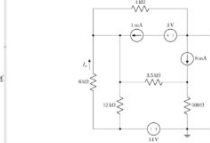
Problem 7

- Determine the number of nodes in the following circuit.



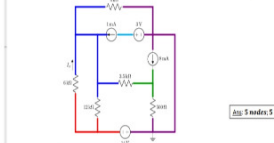
Problem 8

- Determine the number of nodes in the following circuit.



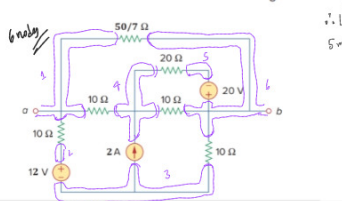
Problem 8: solution

- Question: Determine the number of nodes and meshes in the following circuit.



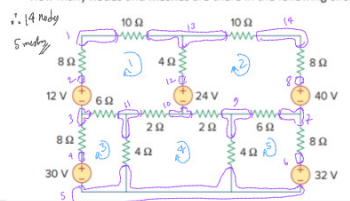
Problem 9

- Determine the number of nodes in the following circuit.



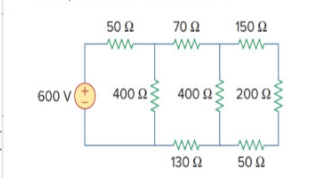
Problem 10

- How many nodes and meshes are there in the following circuit.



Problem 11

- Using series/parallel resistance combination, find the equivalent resistance seen by the source in the circuit below.

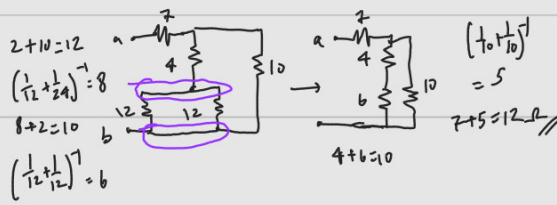
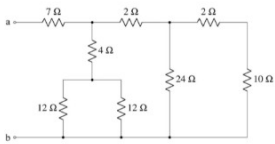


$150 + 200 + 50 = 400 \Omega$
 $\left(\frac{1}{400} + \frac{1}{400}\right)^{-1} = 200$
 $200 + 150 + 50 = 400$
 $\left(\frac{1}{400} + \frac{1}{400}\right)^{-1} = 200$
 $200 + 50 = 250 \Omega$

Ans: R_{eq}

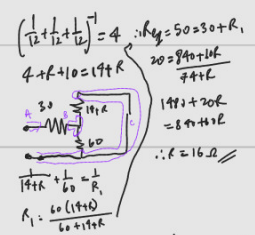
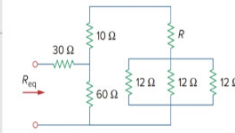
Problem 12

- Find the equivalent resistance between terminals a and b .



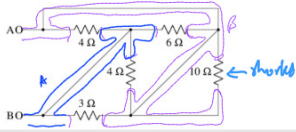
Problem 13

- If $R_{eq} = 50 \Omega$ in the circuit, find R .



Problem 14

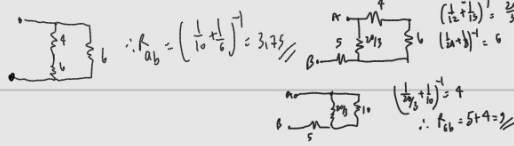
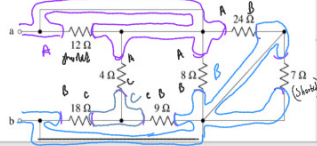
- Find the equivalent resistance between terminals A and B .



All parallel with each other
 $(\frac{1}{4} + \frac{1}{6} + \frac{1}{4} + \frac{1}{3})^{-1} = 1 \Omega$

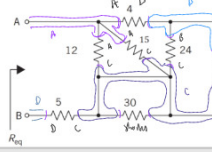
Problem 15

- Find the equivalent resistance between terminals a and b .



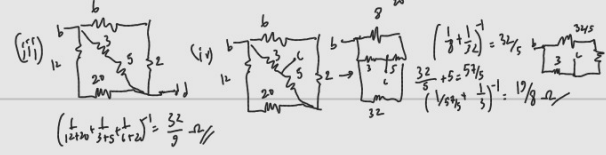
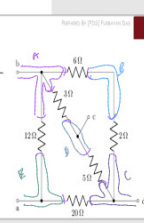
Problem 16

- Determine R_{eq} .



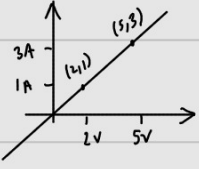
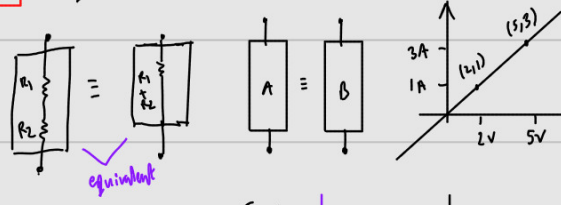
Problem 17

- Find the equivalent resistance between:
 I. a and b ,
 II. a and d ,
 III. b and d , and
 IV. a and c .



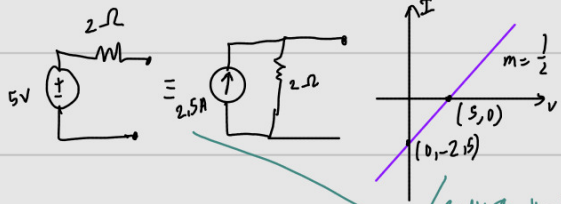
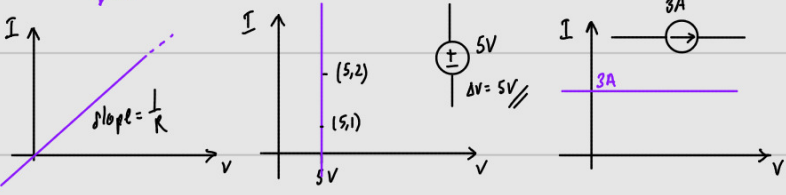
L-3, 4, 5

Equivalent Circuit

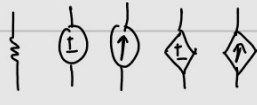


Here I-V characteristics of A & B are same. Thus A & B are equivalent.

In this course, we'll deal with Linear devices only. These give straight line I-V characteristics

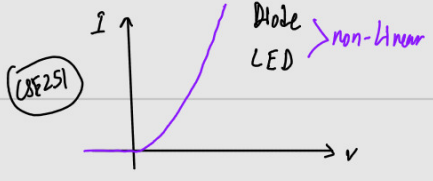


Recall: The direction is opposite of positive sign convention.

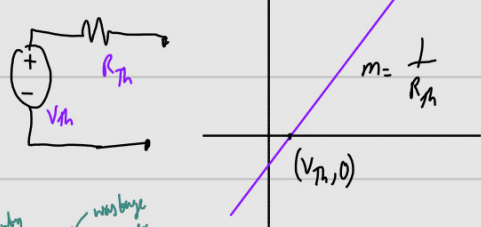
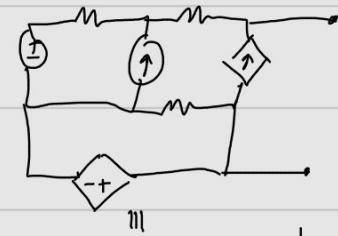


Dependent sources will also give a st. line.

Not necessarily a positive slope every time



Logic gates are made using non-linear circuits

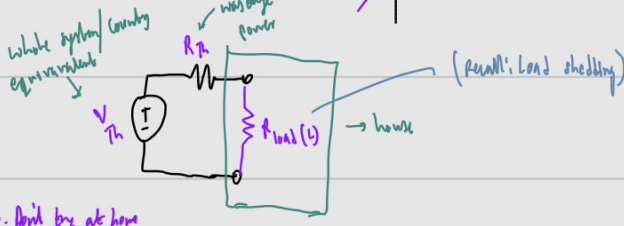
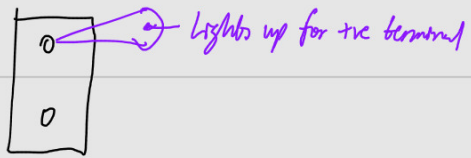


Theremin's Theorem

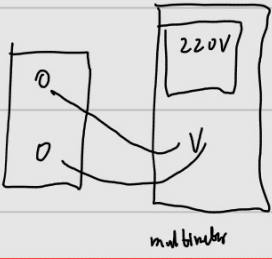
According to Theremin any complicated (Linear) circuit can be reduced and be equivalent to a voltage source and a resistor in series

[Practically]

Source Testing

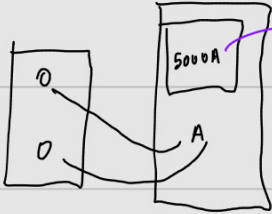


(Recall: Load shedding)



$V_{Th} = 220V$
Open circuit

multimeter



$I_s = 5000A$
Short circuit

Dangerous. Don't try at home

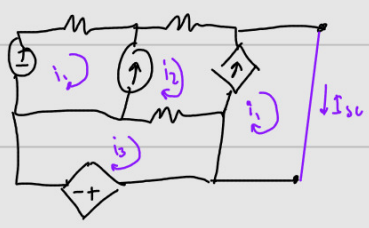
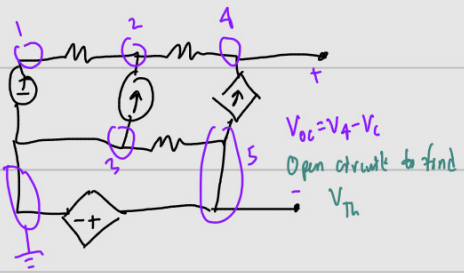
$$\therefore R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{220V}{5000} = 0.044 \Omega$$

This is not the resistance of the house but of the whole system (eg whole BD)

There is no open circuit current, no short circuit voltage

Voltmeter works in open circuit mode

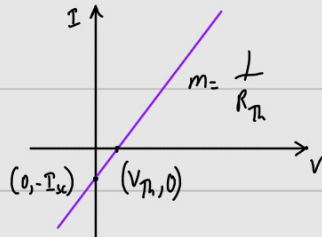
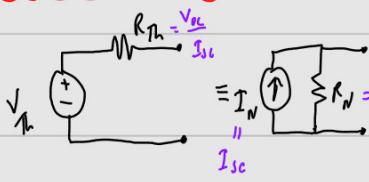
Ammeter works in short circuit mode



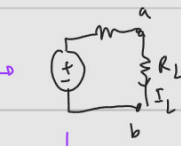
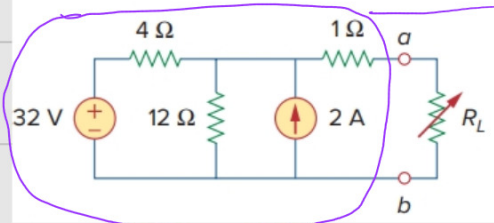
$$V_{Th} = V_{oc}$$

$$R_{Th} = \frac{V_{oc}}{I_{sc}}$$

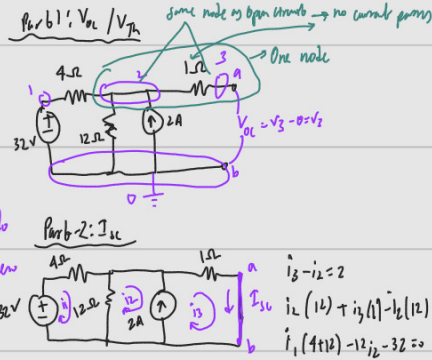
Norton's Theorem



Example 4.8



Thus, we don't need to do mesh analysis every time for new values of R_L

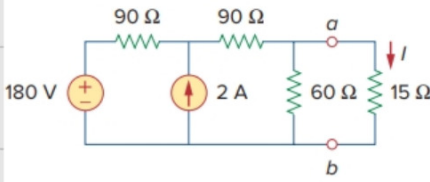


$$V_2 \left(\frac{1}{4} + \frac{1}{12} + \frac{1}{1} \right) - \frac{32}{4} - \frac{V_2}{1} = 0$$

$$V_2 = V_{oc}$$

$$\therefore V_{th} = V_{oc} = 30V$$

Practice Problem 4.8

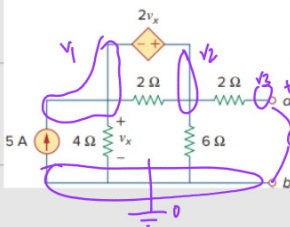


$$V_{th} = 90V, R_{th} = 45\Omega$$

$$I = 1.5A$$

Practice

Example 4.9



Part 1: V_{oc}

$$V_2 = V_3$$

Node 1: V_1

$$V_2 - V_1 = 2V_1$$

$$V_A = V_1 \text{ (Node 1, 2)}$$

$$\therefore V_2 - V_1 = 2V_1$$

$$3V_1 = V_2$$

$\therefore$ solving these eqns

$$V_1 = 6.67V$$

$$V_2 = 20V$$

$$V_3 = 20V$$

$$\therefore V_{oc} = V_3 = 20V = V_{th}$$

Part 2: I_{sc}

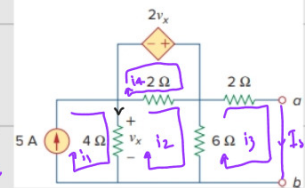
Super node 1, 2

$$V_1 \left(\frac{1}{4} + \frac{1}{2} \right) - \frac{0}{4} - \frac{V_2}{2} - 5 + V_2 \left(\frac{1}{2} + \frac{1}{6} + \frac{1}{2} \right) - \frac{V_2}{2} - \frac{V_2}{2} = 0$$

no mesh. As common in super node

$$\text{Node 3: } V_3 \left(\frac{1}{2} \right) - \frac{V_2}{2} = 0$$

Example 4.9



$$i_1 = 5A, i_2 = 4.44A, i_3 = 3.33A, i_4 = 6.67A$$

$$\therefore R_{th} = \frac{V_{oc}}{I_{sc}} = \frac{20}{3.33} = 6\Omega$$



$$I_{sc} = i_3$$

Step 1: 4 nodes. No supernode

Step 2: mesh: 1, 3

Step 3/4:

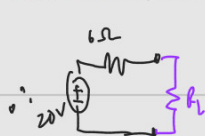
$$\text{Mesh 1: } i_2(4+2+6) - 4i_1 - 2i_3 = 0$$

$$\text{Mesh 3: } i_3(6+2) - 6i_2 = 0$$

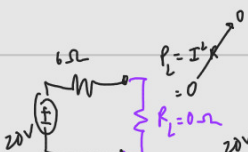
$$\text{Mesh 4: } i_4(2) - 2i_3 - 2V_x = 0$$

$$V_x = (i_1 - i_2) \times 4$$

Maximum Power Transfer Theorem



happens when $R_L = R_{th}$



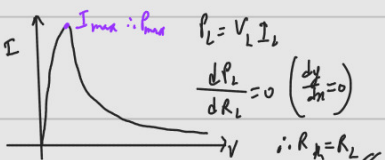
$$P_L = I_L^2 R_L$$

$$P_L = 0$$

$$P_L = 0$$

$$I = 0$$

So more current is somewhere in the $R = 0$ & $R = \infty$. This is when $R_L = R_{th}$



$$P_L = V_L I_L$$

$$\frac{dP_L}{dR_L} = 0 \left(\frac{dV_L}{dR_L} = 0 \right)$$

$$\therefore R_L = R_{th}$$

$$P_L = V_L I_L$$

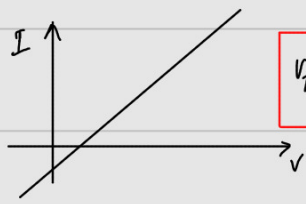
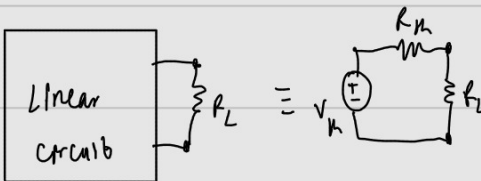
$$V_L = V_{th} \times \frac{R_L}{R_{th} + R_L}$$

$$I_L = \frac{V_{th}}{R_{th} + R_L}$$

12/01/2025
Sunday

Thevenin's Theorem (+ Norton's Theorem)

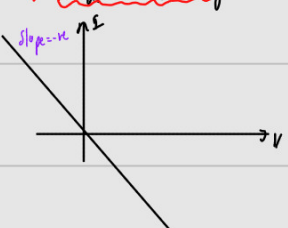
We are able to do equivalent circuit of complicated circuit because they are linear and have same characteristic eqn



$$V_{th} = V_{open\ circuit}, R_{th} = \frac{V_{oc}}{I_{sc}}$$

Maximum Power Transfer Condition
 $R_{th} = R_L$

R_{th} can be negative



$$P = I^2 R = \frac{V^2}{R}$$

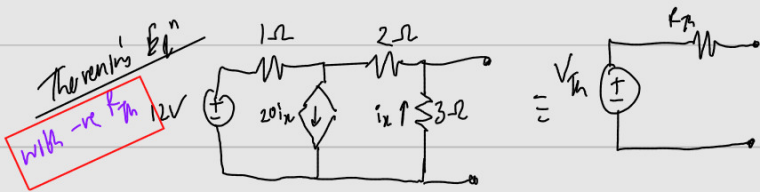
$$= \frac{(V)(I)}{(I)(R)} = \frac{(+ve)}{(+ve)} = (+ve)$$

$$= \frac{(V)(I)}{(I)(R)} = \frac{(+ve)}{(-ve)} = (-ve)$$

(Power consuming)

(Power supplying)

In real life $R = 0\Omega$ and $R = -ve\Omega$ is not possible



Example 1 (Thevenin)

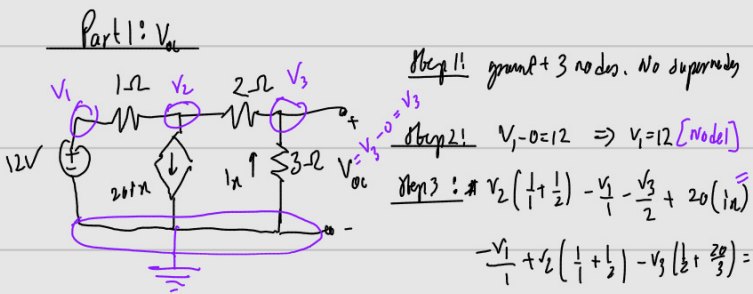
Steps to solve in this Thevenin example:

- 1. V_{oc} will be negative if i_x is negative. Although usually the potential V means the voltage of a point above and across some path, it is not in our case. But we can still make the assumption that V_{oc} is positive.
- 2. Assume a test source of $1A$ which forces i_x to be $1A$. Then we can solve for V_{oc} using KVL.
- 3. KVL: $12V - 1\Omega \cdot 1A - 20\Omega \cdot i_x - 2\Omega \cdot 1A - 3\Omega \cdot i_x = 0$
- 4. $12 - 1 - 20i_x - 2 - 3i_x = 0 \Rightarrow 9 - 23i_x = 0 \Rightarrow i_x = \frac{9}{23}A$
- 5. $V_{oc} = 12V - 1\Omega \cdot 1A - 20\Omega \cdot \frac{9}{23}A - 2\Omega \cdot 1A - 3\Omega \cdot \frac{9}{23}A = -2.571V$

Example 1 (I-V/Step 3)

Steps to solve in this I-V characteristic example:

- 1. Now, we test the circuit with the test source $1A$ and solve for V_{oc} using KVL.
- 2. KVL: $12V - 1\Omega \cdot 1A - 20\Omega \cdot i_x - 2\Omega \cdot 1A - 3\Omega \cdot i_x = 0$
- 3. $12 - 1 - 20i_x - 2 - 3i_x = 0 \Rightarrow 9 - 23i_x = 0 \Rightarrow i_x = \frac{9}{23}A$
- 4. $V_{oc} = 12V - 1\Omega \cdot 1A - 20\Omega \cdot \frac{9}{23}A - 2\Omega \cdot 1A - 3\Omega \cdot \frac{9}{23}A = -2.571V$



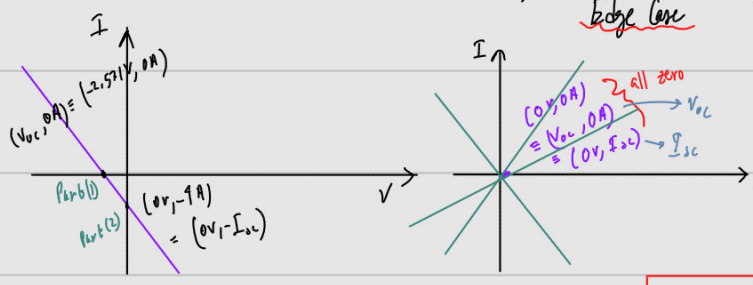
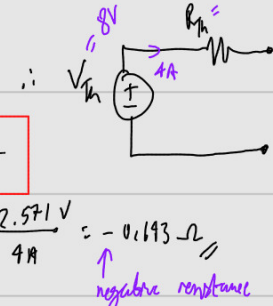
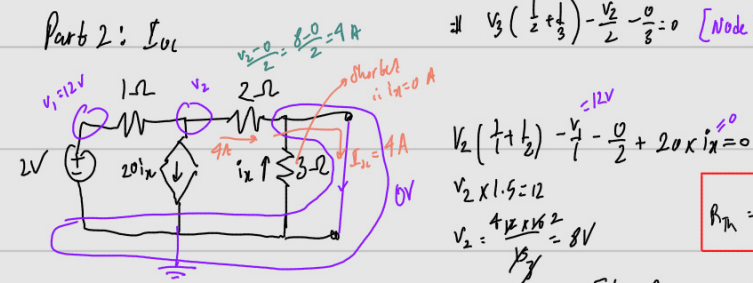
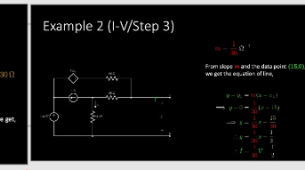
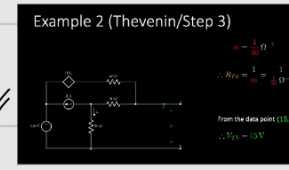
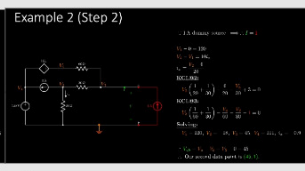
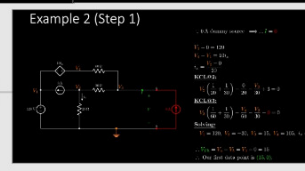
Solving V_{oc} :

$$V_1 = 12V$$

$$V_2 = -4.286V$$

$$V_3 = -2.571V$$

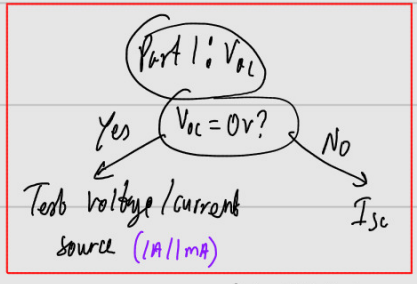
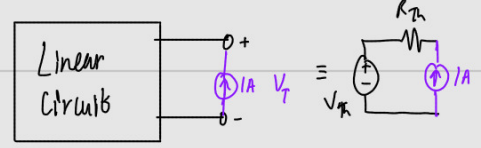
$$V_{oc} = -2.571V$$



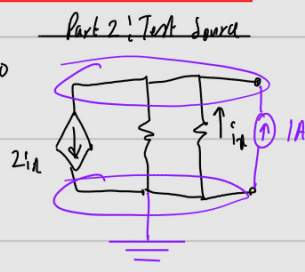
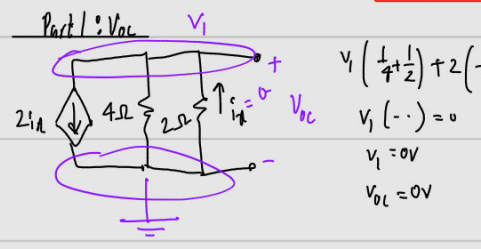
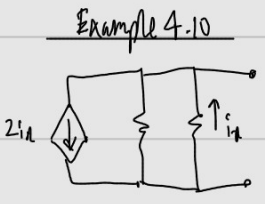
$R_{Th} = \frac{V_{oc}}{I_{sc}} = \frac{0}{0}$

When we get V_{oc} or I_{sc} zero, the other will definitely be zero. So use another data points.

Better use $1A$. Easy to calculate.



$V \times A = W$



$V_1 \left(\frac{1}{4} + \frac{1}{2} \right) + 2 \left(-\frac{V_1}{2} \right) = 0$

$V_1 = 4V$

$V_1 = V_1 - 0 = 4V$

$R_{Th} = \frac{V_1}{I_1} = \frac{4V}{1A} = 4\Omega$

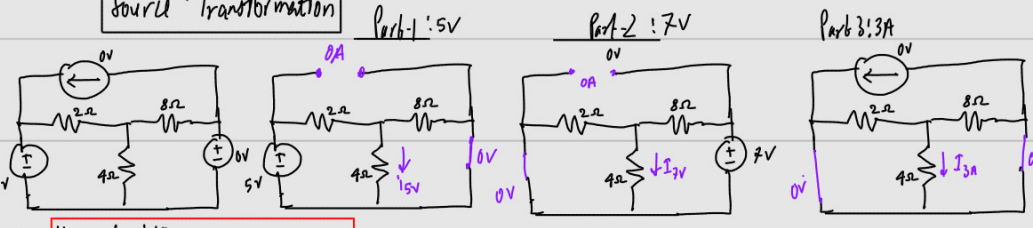
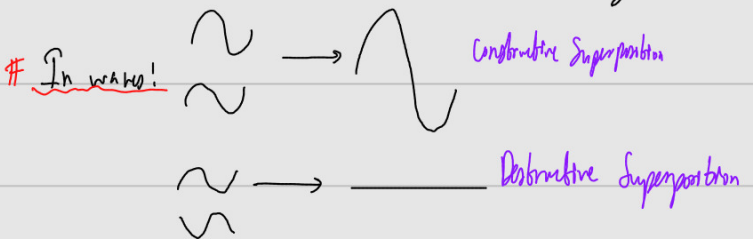
13/01/2025
Monday

Superposition Theorem Only with linear elements L-14

Tools

- Node Analysis
- Mesh Analysis
- Source Transformation

Theremin, Superposition
Transient Analysis, AC Analysis
Capacitor, Inductor



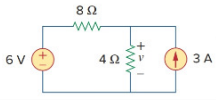
Superposition is mainly imp when using AC source. Sources might have different frequency

Fact: USA, Canada uses 60Hz, UK uses 50Hz

How to kill
→ Current source → Open
→ Voltage source → Short

Chapter 4.3: Superposition Theorem. Example 4.3

Example 4.3



Part 1: 6V



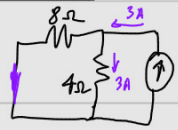
$$i_1(8+4) - 6 = 0$$

$$i_1 = 0.5A$$

$$i_{6V} = -i_1 = -0.5A$$

$$V_{6V} = 4i_1 = (4 \times 0.5)V = 2V$$

Part 2: 3A



$$23A = 3A \times \frac{4}{4+8} = 3A \times \frac{4}{12} = 1A$$

$$V_{3A} = 2A \times 4\Omega = 8V$$

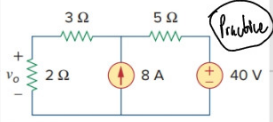
Very Superposition Theorem,

$$V = V_{6V} + V_{3A} = 2V + 8V = 10V$$

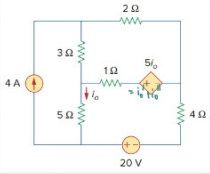
$$I = I_{6V} + I_{3A} = -0.5 + 1 = 0.5A$$

Do not kill the dependent sources

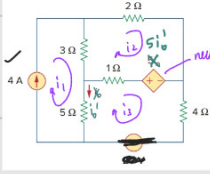
Practice Problem 4.3



Example 4.4



Example 4.4



$i_1 = 4A$ (method 1)

$i_2(3+2+1) - i_3 - 5i_1 = 0$ (method 2)

$i_1' = i_1 - i_3$

$i_3(5+1+4) - 5i_1 - i_2 + 5i_1 = 0$ (method 3)

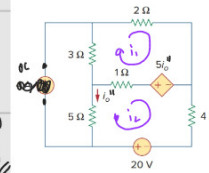
$i_1 = 4A$

$i_2 = 4.706A$

$i_3 = 0.941A$

$i_1' = 4 - 0.941 = 3.059A$

Example 4.4



$i_1' = i_1 - i_3$

$i_1(3+2+1) - i_2 - 5i_1' = 0$ (method 1)

$i_2(5+1+4) - i_1 + 5(-i_2) - 20 = 0$ (method 2)

$i_1 = -2.353A$

$i_2 = 3.534A$

$i_1' = -i_2 = -3.534A$

$i_1' = i_1 - i_3$

$i_3 = 3.06A + (-3.534A)$

$i_3 = -0.474A$



8) what is the node voltage?

= no voltage, no current flows in no node voltage



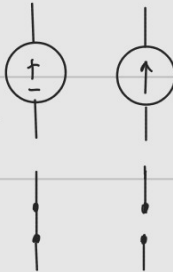
As no independent source

by dependent source

which acts like resistor

Summary

- kill only independent source
- How to kill:



Superposition Theorem can be applied to find

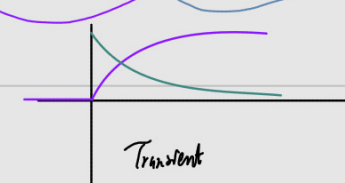
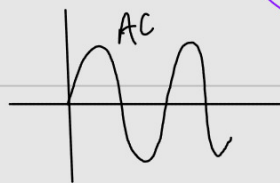
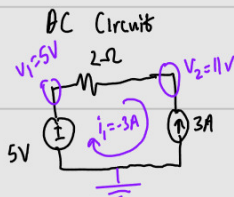
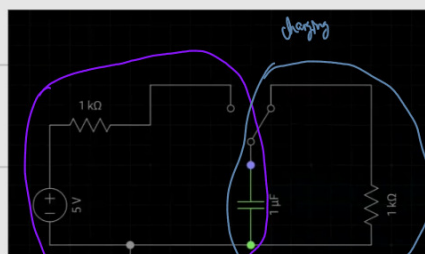
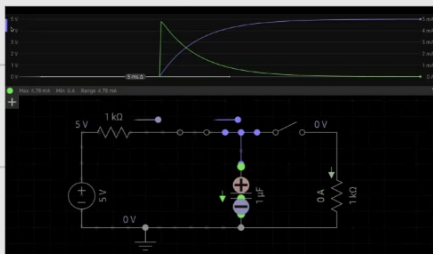
- Voltage ✓ $P \neq P_{A1} + P_{A2}$
- Current ✓ $I^2 \neq I_{A1}^2 + I_{A2}^2$
- Power ✗ $(I_{A1} + I_{A2})^2 \neq I_{A1}^2 + I_{A2}^2$

Online class

RC Circuit Introduction

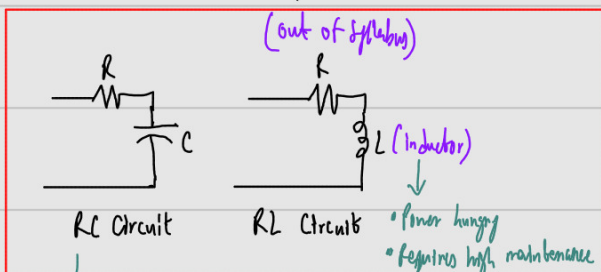
20/12/2024

For day



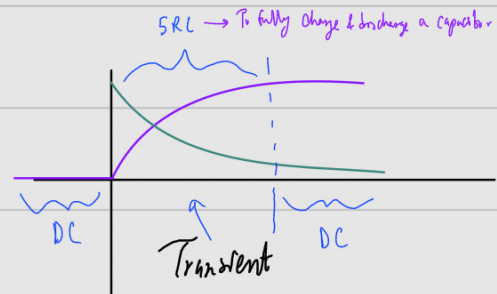
A lot of transistors circuit is used inside RAM

In fact every bit is a transistors circuit: resistance & capacitor



Speed at which CPU can work is determined by the values of R and C.

* 56 He → One of the fastest in today's time

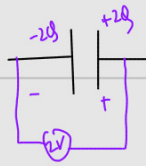
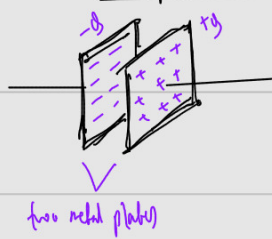


$$5RC = 5 \times 1k\Omega \times 1\mu F$$

$$= 15ms$$

To fully charge & discharge 1μF

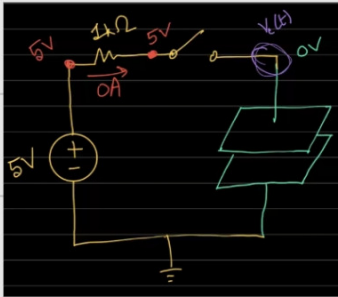
Capacitor



Capacitor wants to be an open circuit but takes time to open. This time is transient behaviour

$$Q = CV$$

- ↳ Capacitor finally becomes open
- ↳ Capacitor's voltage always changes gradually



↳ DC Analysis

↳ Transient Analysis

↳ AC Analysis

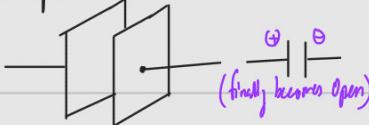
Capacitor, Inductor

Transient → Temporary

L-15

14/01/2025
Tuesday

Capacitor



$$V = RI$$

resistor resists current

$$Q = CV$$

$$\Rightarrow \frac{dQ}{dt} = C \frac{dV}{dt}$$

$$I = C \frac{dV}{dt}$$

$$\frac{e}{dt} = \frac{e}{dt}$$

C [For μF]

capacitor resists the change in voltage

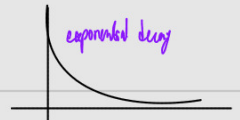
Inductor

(finally becomes short circuit)
 ΔI Unit: H (Henry)

*** Whenever current through Inductor is changed, a voltage difference is created

$$\text{Here } V \propto \frac{dI}{dt}$$

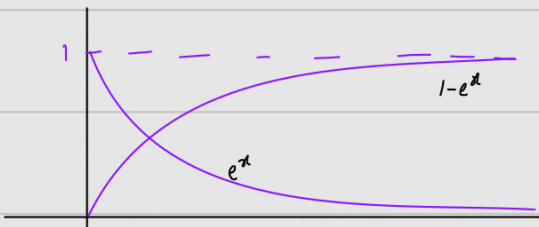
$$V = L \frac{dI}{dt}$$



Inductance resists change in current

*** Inductor tries to hold current and Capacitor tries to hold voltage

*** Capacitor basically works like a battery. When we charge a battery, it charges quickly in the beginning, but slows down when close to 95% - 100%.



$$V(t) = (V_f - V_i)(1 - e^{-t/\tau}) + V_i$$

$$\tau = RC$$

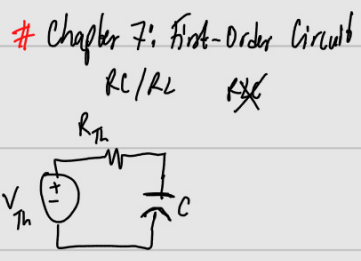
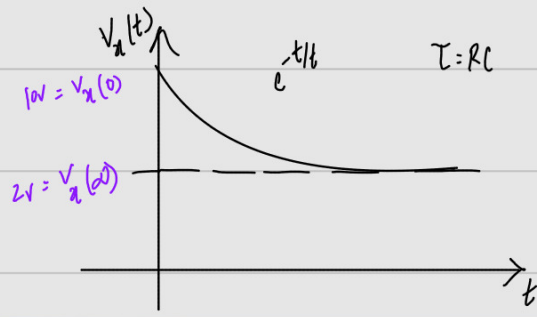
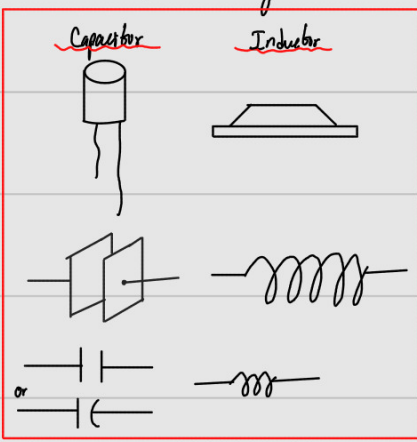
$$\therefore V(t) = (V_f - V_i)(1 - e^{-t/\tau}) + V_i$$

$$I(t) = (I_f - I_i)(1 - e^{-t/\tau}) + I_i$$

$$\tau = \frac{L}{R} \quad I_s = \frac{1H}{1\Omega}$$

Transient Analysis

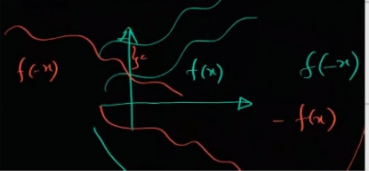
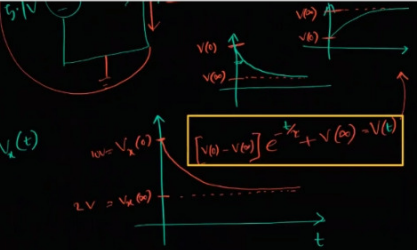
L-16



Practice Problem 7.1 Refer to the circuit in Fig. 7.7. Let $v_C(0) = 60$ V. Determine v_C , v_x , and i_x for $t \geq 0$.

Answer: $60e^{-0.25t}$ V, $20e^{-0.25t}$ V, $-5e^{-0.25t}$ A.

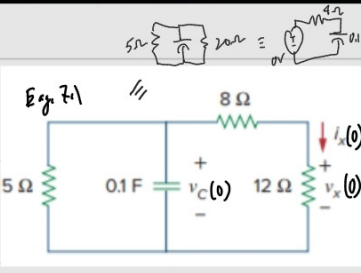
Figure 7.7 For Practice Prob. 7.1.



$$V_C(t) = V_C(\infty) + [V_C(0) - V_C(\infty)]e^{-t/\tau}$$

$$i_x(t) = i_x(\infty) + [i_x(0) - i_x(\infty)]e^{-t/\tau}$$

$$V_x(t) = V_x(\infty) + [V_x(0) - V_x(\infty)]e^{-t/\tau}$$



Initial Circuit

In this circuit, 5Ω 8Ω 12Ω $0.1F$ $15V$

Let $V_C(0) = 15V$

Find V_C , V_x , i_x

$V_C(0) = 15V$

$i_x(0) = \frac{15V}{8 + 12} = 0.75A$

$V_x = 12 \times i_x(0) = 12 \times \frac{3}{4} = 9V$

Final Circuit

after infinite time

Complete Eqn

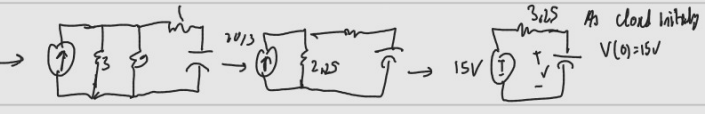
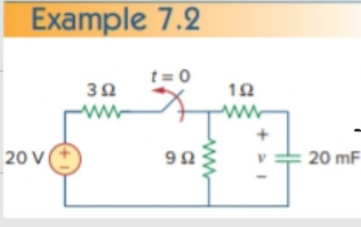
$$V_C(t) = [V_C(0) - V_C(\infty)]e^{-t/\tau} + V_C(\infty)$$

$$= [15 - 0]e^{-t/0.4} + 0$$

$$= 15e^{-2.5t} V$$

$$V_x(t) = V_x(\infty) + [V_x(0) - V_x(\infty)]e^{-t/\tau}$$

$$= 0 + [9 - 0]e^{-t/0.4}$$

$$= 9e^{-2.5t} V$$


replace capacitor/inductor with proper element

Initial Circuit $\rightarrow x(0) \rightarrow$ actually $x(0)$

Intermediate Substitution $\rightarrow \tau = R_{th}C / \frac{L}{R_{th}}$

Final Circuit $\rightarrow x(\infty)$

$$x(t) = [x(\infty) - x(0)]e^{-\frac{t-0}{\tau}} + x(0)$$

Initial Circuit $\rightarrow x(0)$

Intermediate Circuit

Final Circuit

$V(t) = 0 + [15 - 0]e^{-t/0.2} + 0$

$= 15e^{-5t} V$

$\tau = R_{th}C = 10 \times 20 \times 10^{-3} = 0.2s$

L-17

Capacitor stores energy as electrical energy $\rightarrow$ Electrostatic energy

Inductor stores energy magnetically $\rightarrow$ Magnetic energy

Energy stored initially $= \frac{1}{2} C V(0)^2$

(Capacitor) $\frac{1}{2} C V^2 = \frac{1}{2} \times 20mF \times (15V)^2 = 2.25J$

Energy stored $= \frac{1}{2} L I^2$

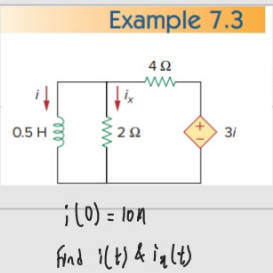
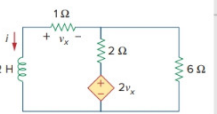
(Inductor) $\frac{1}{2} L I^2$

Practice Problem 7.3

Find i and v_x in the circuit of Fig. 7.15. Let $i(0) = 7$ A.

Answer: $7e^{-2t}$ A, $-7e^{-2t}$ V, $t > 0$.

Probleme



Initial

Intermediate

Final

$i(0) = 10A$

Find $i(t)$ & $i_x(t)$

Final

$i(t) = [i(0) - i(\infty)]e^{-t/\tau} + i(\infty)$

$i(t) = [10 - 0]e^{-t/0.5} + 0$

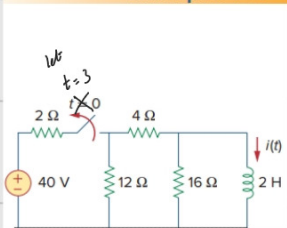
$= 10e^{-2t} A$

$i_x(t) = [i_x(0) - i_x(\infty)]e^{-t/\tau} + i_x(\infty)$

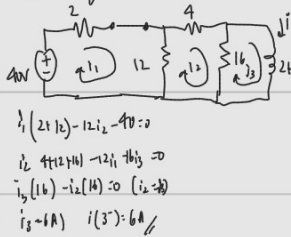
$i_x(t) = [-1.67 - 0]e^{-t/0.5} + 0$

$= -1.67e^{-2t} A$

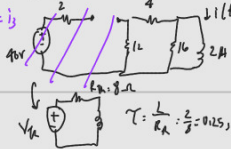
Example 7.4



Initially



Intermediate



Final

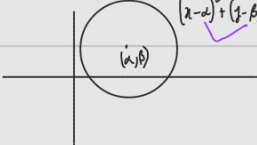
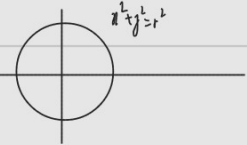
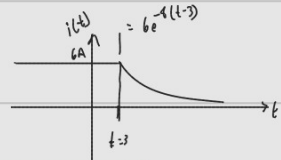


By

$$i(t) = [i(3^-) - i(\infty)] e^{-(t-3)/\tau} + i(\infty)$$

$$= [6 - 0] e^{-\frac{t-3}{0.667}} + 0$$

$$= 6e^{-1.5(t-3)}$$



Max Power Transferred to the load → When $R_L = R_{Th}$

$R_{Th} = 0.967\Omega$

$V_{Th} = 1.984V$

$R_{load} = 0.967\Omega = R_{Th}$

$P = \left(\frac{V_{Th}}{2R_{Th}} \right)^2 \times R_{Th} = \frac{V_{Th}^2}{4R_{Th}}$

1, 2, 3, 4, ... → $\mathbb{N}$ (Natural numbers)

... -2, -1, 0, 1, 2, 3, ... → $\mathbb{Z}$ (Whole numbers)

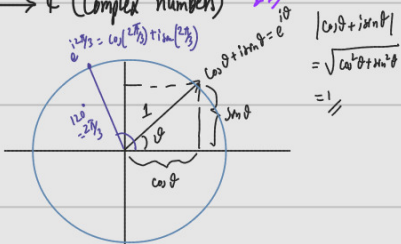
... -8, 0, 1/2, 4, ... → $\mathbb{Q}$ (Rational numbers)

... -1/2, 0, √2, √3, π, e → $\mathbb{R}$ (Real numbers)

... -√3i, i, 2i, 2.5i ... → Imaginary numbers

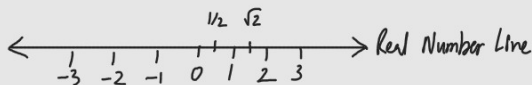
... √2 - √3i → $\mathbb{C}$ (Complex numbers)

$$e^{j\theta} = \cos\theta + j\sin\theta$$

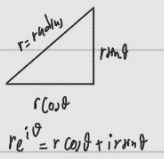
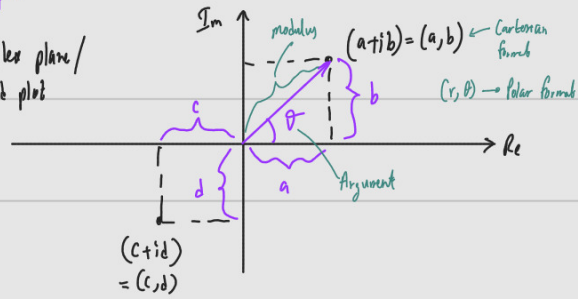


L-18

$a+bi$
real part → a
imaginary part → bi



Complex plane / Argand plot

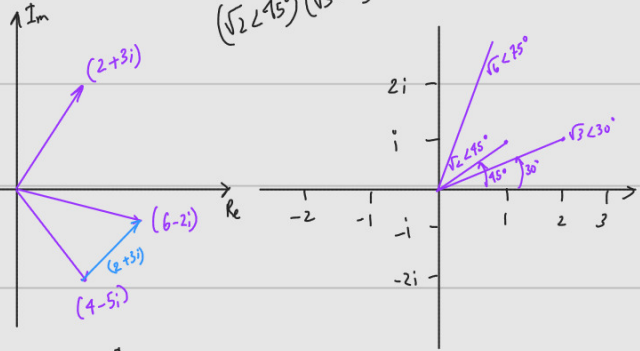
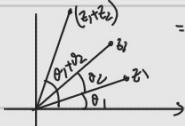


$$z_1 = r_1 e^{j\theta_1} = r_1 \angle \theta_1$$

$$z_2 = r_2 e^{j\theta_2} = r_2 \angle \theta_2$$

$$z_1 z_2 = (r_1 e^{j\theta_1}) (r_2 e^{j\theta_2}) = (r_1 r_2) e^{j(\theta_1 + \theta_2)}$$

$$z_1 / z_2 = r_1 / r_2 e^{j(\theta_1 - \theta_2)}$$

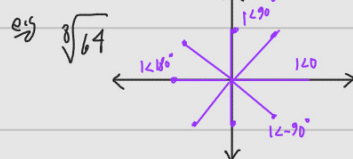
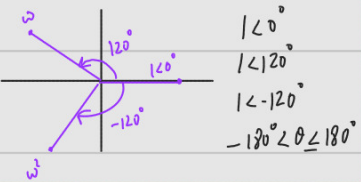


$$\sqrt[3]{1} = \omega = 1 \angle 120^\circ$$

$$\omega^3 = \omega \times \omega \times \omega$$

$$= 1 \angle 120^\circ \times 1 \angle 120^\circ \times 1 \angle 120^\circ$$

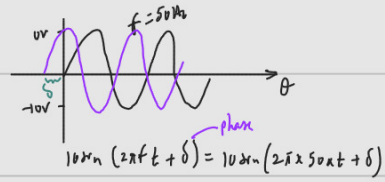
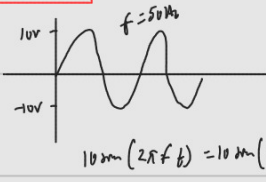
$$= 1 \angle 360^\circ = 1 \angle 0^\circ = 1$$



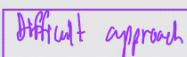
$$e^{j(A+B)} = \cos(A+B) + j \sin(A+B)$$

$$\text{or } e^{jA} \cdot e^{jB} = (\cos A + j \sin A) \cdot (\cos B + j \sin B)$$

$$= \cos A \cos B - \sin A \sin B + j(\cos A \sin B + \sin A \cos B)$$

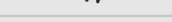


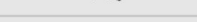
Quarterton
a + b + j + k
e hyper dimension






→ All 3 treated the same

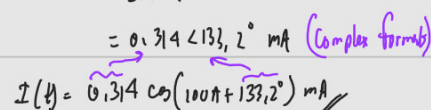



2: $\neq 90^\circ$

$Z = R$
 $= R \angle 0^\circ$

$Z = \frac{1}{j\omega C} = -\frac{j}{\omega C}$
 $= \frac{1}{\omega C} \angle -90^\circ$

$Z = j\omega L$
 $= \omega L \angle 90^\circ$

$$1: a+b \quad 2: r < 0$$

$$\sin(\theta + 90^\circ) = -\cos\theta$$

$$\cos\theta = \sin(90^\circ - \theta)$$

$$Z_{23} = \frac{Z_2 \cdot Z_3}{Z_2 + Z_3} = \frac{25\angle -30^\circ \cdot 20\angle 90^\circ}{25\angle -30^\circ + 20\angle 90^\circ}$$

$$= 100\angle 90^\circ \Omega //$$

$$V_o(t) = V \times \frac{Z_{23}}{Z_1 + Z_{23}} = 1$$

$$V_o(t) = 1$$

$v_1 \left(\frac{1}{4} + \frac{1}{3L} \cdot 90^\circ \right) - \frac{v_2}{4} = \frac{0}{4} - 3L \cdot 0 + v_2 \left(\frac{1}{4} + \frac{1}{6L} \cdot 90^\circ + \frac{1}{12} \right) - \frac{v_1}{4}$
 $v_1 \cdot 0.33 \angle 90^\circ + v_2 \cdot 0.186 \angle -63.4^\circ = 3L \cdot 0$
 can't solve in calculator (cassio). so solve without OR

$$r_1 \angle \theta_1 \times r_2 \angle \theta_2 = r_1 r_2 \angle (\theta_1 + \theta_2)$$

$$V_2 = 31.7 \angle -86.7^\circ$$

$$V_1 = V_1 + 10 \angle 45^\circ = 26.2 \angle -67.7^\circ$$

$$\begin{array}{l} a_1 x + b_1 y = c_1 \\ a_2 x + b_2 y = c_2 \end{array} \quad x = \frac{\begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}} \quad y = \frac{\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}}{\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}}$$

$$V_1 \cdot 0.33 \angle 20^\circ + V_2 \cdot 0.11 \angle -63.1^\circ = 3 \angle 0^\circ$$

$$V_1 - V_2 = 10 \angle 45^\circ$$

$$V_1 = \frac{\begin{vmatrix} 10 \angle 45^\circ & -1 \\ 3 \angle 6^\circ & 0.1186 \end{vmatrix}}{\begin{vmatrix} 1 & -1 \\ 0.133 \angle 9^\circ & 0.1186 \angle -63.4^\circ \end{vmatrix}} = \frac{10 \angle 45^\circ \times 0.1186 \angle -63.4^\circ - 3 \angle 6^\circ \times 1}{1 \times 0.1186 \angle -63.4^\circ - (-1) \times 0.133 \angle 9^\circ} \\ \approx 26.14 \angle -70^\circ$$

$$a_1x + b_1y + c_1z = d_1$$

$$a_2x + b_2y + c_2z = d_2$$

$$a_3x + b_3y + c_3z = d_3$$

$$J = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

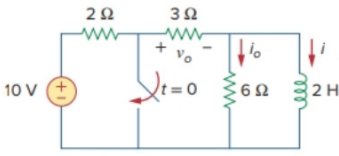
$$\begin{array}{|c|c|c|} \hline a_1 & b_1 & c_1 \\ \hline a_2 & b_2 & c_2 \\ \hline a_3 & b_3 & c_3 \\ \hline \end{array}$$

$$= a_1 (b_2 c_3 - b_3 c_2) - b_1 (a_2 c_3 - a_3 c_2) + c_1 (a_2 b_3 - a_3 b_2)$$



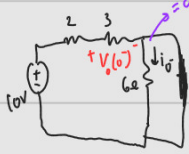
$$i_3 = 5 \angle 0^\circ \text{ [mA]} \\ i_1 (8 + 10 \angle 90^\circ + 2 \angle -90^\circ) - i_3 (10 \angle 90^\circ) - i_2 (2 \angle -90^\circ) = 0 \\ i_2 (4 + 2 \angle -90^\circ + 2 \angle -90^\circ) - i_1 (2 \angle -90^\circ) - i_3 (2 \angle 90^\circ) + 2 \angle 90^\circ = 0$$

Example 7.5

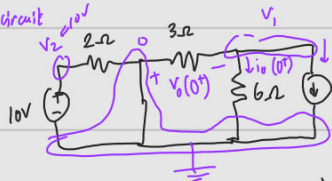


$$i(t) = [i(0^+) - i(\infty)] e^{-\frac{t}{\tau}} + i(\infty) \\ i(0^+) = i(0^-) \rightarrow \text{same for Inductor}$$

$t = 0^- (t < 0)$



$$i_o(0^-) = 0A \\ i(0^-) = \frac{10}{2+3} = 2A \\ V_o(0^-) = 10 \times \frac{3}{2+3} = 6V$$

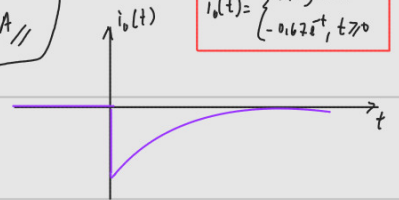


$$V_1 \left(\frac{1}{6} + \frac{1}{3} \right) - \frac{0}{6} - \frac{0}{3} + 2 = 0 \quad | \quad i_o(0^+) = 0A \\ V_1 = \frac{-2}{\frac{1}{6} + \frac{1}{3}} = -4 \\ V_o(0^+) = 0 - V_1 = 4V, \quad i_o(0^+) = \frac{V_1}{1\Omega} = -0.67A$$

$\tau = 1s$

$$i(t) = [i(0^+) - i(\infty)] e^{-\frac{t}{\tau}} + i(\infty) \\ = [-0.67 - 0] e^{-t} + 0 \\ = -0.67 e^{-t} A //$$

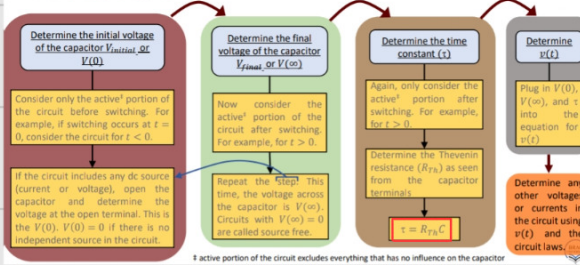
$$i_o(t) = \begin{cases} 0A & t < 0 \\ -0.67A & t \geq 0 \end{cases}$$



[500] YT

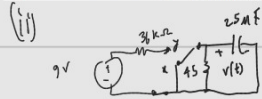
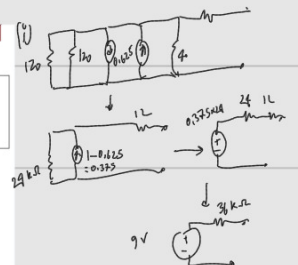
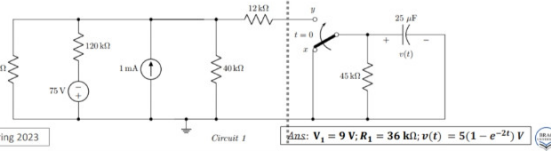
Transient Analysis Problem Solving

Procedure

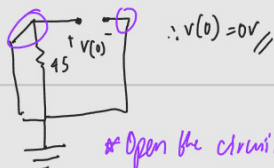


Problem 14

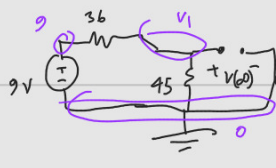
- Reduce the left portion with respect to the dashed grey line of Circuit 1 so that it takes the form of Circuit 2 as shown. Write down the values of V_1 and R_1 .
- Now, analyse the Transient Behaviour of the circuit assuming that the switch moves from position x to position y at $t = 0$. Determine $v(t)$ for $t > 0$.



1] Initial circuit $\xrightarrow{\text{Capacitor open}} V_c(0^-)$ 2] Final circuit $\xrightarrow{\text{Capacitor open}} V_c(\infty)$ 3] Time constant τ

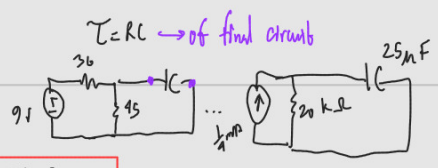


$$V_c(0^-) = 0V //$$



$$V_1 = 5V //$$

$$V_c(\infty) = 5V //$$



$\tau = R_{Th} \times C \rightarrow \text{of final circuit}$

$$\tau = R_{Th} \times C = 20k\Omega \times 25\mu F = 0.5s //$$

$$\text{Full charge: } 5V = 5 \times 0.5 = 2.5s //$$

$$V(t) = \begin{cases} V_c(0^-) & t < 0 \\ V_c(\infty) + [V_c(0^-) - V_c(\infty)] e^{-t/\tau} & t \geq 0 \end{cases}$$

$$V(t) = \begin{cases} 0V; & t < 0 \\ 5 + [0 - 5] e^{-t/0.5} & t \geq 0 \end{cases} \quad \text{when } t = 5\tau$$

$$V(\infty) = 5 - 5 e^{-2(\infty)} = 5V$$

$$V(1) = 5 - 5 e^{-2(1)} = 4.323V$$

$$V(2.5) = 4.946V \approx 99.9\%$$

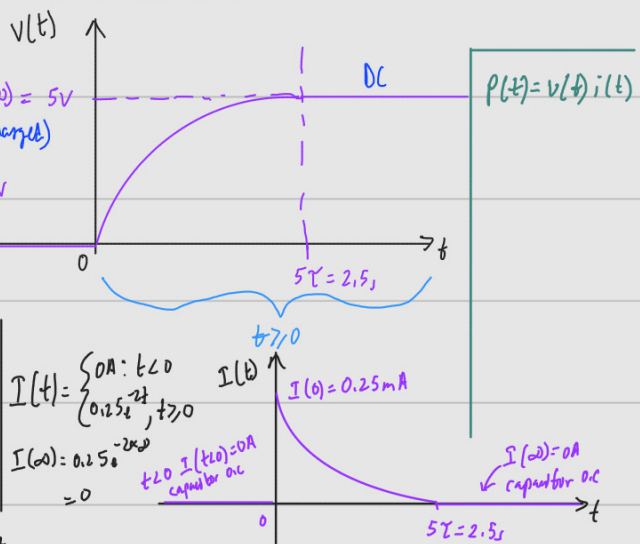
(almost 5)

$$I(t) = C \frac{dV}{dt}$$

$$I(t) = C \frac{dV}{dt} \quad t < 0 \quad I(t) = 0A$$

$$I(t) = C \frac{dV}{dt} \quad t \geq 0 \quad I(t) = C \frac{d}{dt} [5 - 5e^{-2t}]$$

$$= C \left[0 - (-5) \cdot (-2) e^{-2t} \right] = -10 \times 25 \times 10^{-6} e^{-2t} A = -0.25 \times 10^{-3} e^{-2t} A = -0.25 e^{-2t} \text{ mA} //$$



Initial and final current of capacitor is always zero except for transient period. At rest of the time it acts as an open circuit when it's stable.

Final Review (Fall 2024)

- 1) Transient → No dependent source
- 2) Superposition/Source Transformation → R_k by killing all sources (superposition) and find eq resistance
- 3) I-V → Source Transformation
- 4) AC Sinusoidal

$A \sin(\omega t + \phi)$
 $A > 0$ $\omega > 0$ $-180^\circ \leq \phi \leq 180^\circ$

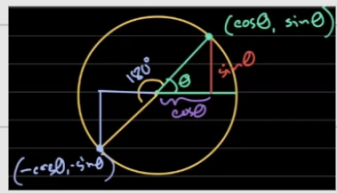
$\sin(\theta - 90) = \cos(\theta - 90)$
 $\sin(\theta + 90) = \cos \theta$
 $\sin(-\theta) = -\sin \theta$
 $\cos(-\theta) = \cos \theta$

Always keep ϕ betⁿ -180 & 180
 (Add/Subtract 360° when needed to adjust)
 $\sin(180 + \theta) = -\sin \theta$
 $\cos(180 + \theta) = -\cos \theta$

Problem 3

- Calculate amplitude, initial phase ($-180^\circ \leq \phi \leq 180^\circ$), angular frequency, period, and frequency for the following sinusoids. What are the values of V_s and I_s at $t = 20$ ms? (Both sin and cosine forms can be employed)
- I. $V_s = 45 \cos(5\pi t + 36^\circ)$ (V)
- II. $I_s = 15 \cos(25\pi t + 25^\circ)$ (A)
- III. $I_s = -20 \cos(314t - 30^\circ)$ (A)
- IV. $V_s = -4 \sin(628t + 55^\circ)$ (V)

Ans: I. 45 V; 36° or 126°; 5π rad/s; 0.4 s; 2.5 Hz; 26.45 V
 II. 15 A; 25° or 155°; 25π rad/s; 80 ms; 12.5 Hz; -6.34 A
 III. 20 A; 150° or 30°; 314 rad/s; 20 ms; 50 Hz; -17.29 A
 IV. 4 V; 125° or 25°; 628 rad/s; 10 ms; 100 Hz; -3.26 V



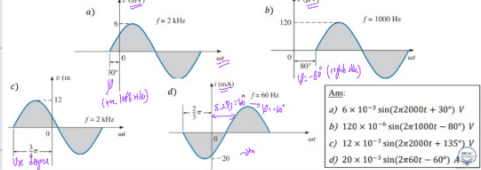
(i) $V_s = 45 \cos(5\pi t + 36^\circ)$ V
 $A = 45$ V, $\omega = 5\pi$, $\phi = 36^\circ$
 OR $V_s = 45 \cos(5\pi t + 36^\circ + 90^\circ)$
 $= 45 \sin(5\pi t + 126^\circ)$

(ii) $I_s = 15 \cos(25\pi t + 25^\circ)$ A
 $A = 15$ A, $\omega = 25\pi$ rad/s, $\phi = 25^\circ$
 OR $I_s = -20 \cos(314t - 30^\circ)$ A
 $= 20 \cos(314t + 150^\circ)$ [cos(180+θ) = -cos θ]
 $A = 20$ A, $\omega = 314$ rad/s, $\phi = 150^\circ$

(iv) $V_s = -4 \sin(628t + 55^\circ)$ V
 $= 4 \sin(628t + 55^\circ + 180^\circ)$
 $= 4 \sin(628t + 235^\circ - 36^\circ)$
 $= 4 \sin(628t - 125^\circ)$
 $\therefore A = 4$ V, $\omega = 628$ rad/s, $\phi = -125^\circ$

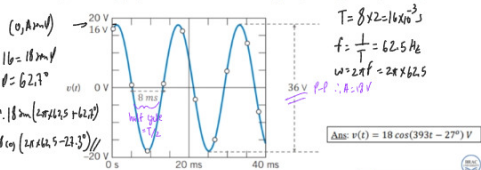
Problem 4

- Write analytical expressions for the waveforms with the initial phase in degrees.



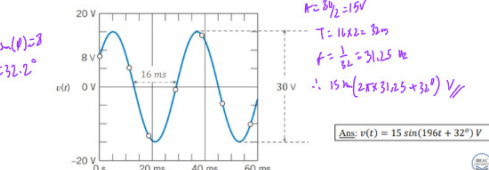
Problem 5

- The following figure shows a sinusoidal voltage $v(t)$, plotted as a function of time t . Represent $v(t)$ by a function of the form $A_m \cos(\omega t + \phi)$ with ϕ in degrees.



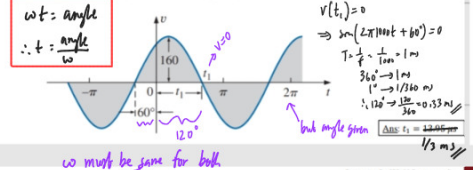
Problem 6

- The following figure shows a sinusoidal voltage $v(t)$, plotted as a function of time t . Represent $v(t)$ by a function of the form $A_m \sin(\omega t + \phi)$ with ϕ in degrees.



Problem 7

- A sinusoidal voltage $v(t) = 160 \sin(2\pi 1000t + 60^\circ)$ is plotted as a function of time t . Determine the time t_1 when the waveform crosses the axis.



Problem 8

- Consider the sinusoidal voltage $v(t) = 80 \cos(1000\pi t - 30^\circ)$ V.

- What is the first time after $t = 0$ that $v(t) = 80$ V?
- If the sinusoidal voltage is shifted $2/5$ ms to the left along the time axis, what will be the expression for $v(t)$?
- What is the minimum number of milliseconds that the function must be shifted to the (right/left) if the expression for $v(t)$ is $80 \sin(1000\pi t)$ V?

Ans:
 (a) $t = 166.67 \mu$ s
 (b) $-80 \sin(1000\pi t)$ V
 (c) $t = 333.33 \mu$ s

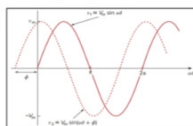
Problem 9

- At $t = -2$ ms, a sinusoidal voltage $v(t)$ is known to be zero and going positive. The voltage is next zero at $t = 8$ ms. It is also known that the voltage is 80.9 V at $t = 0$. What is the expression for $v(t)$?

Ans: $v(t) = 20 \cos(400\pi t - 45^\circ)$ V

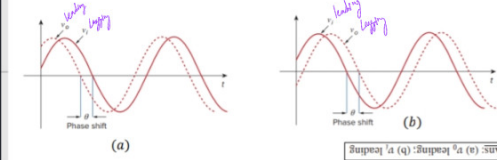
Leading and Lagging Sinusoids

- The equation for a sinusoid in more general form, $v(t) = V_m \sin(\omega t + \phi)$ where $(\omega t + \phi)$ is the argument and ϕ is the initial phase. Both argument and phase can be in radians or degrees.
- Let's examine two sinusoids $v_1(t) = V_{m1} \sin \omega t$ and $v_2(t) = V_{m2} \sin(\omega t + \phi)$
- The starting point of v_2 occurs first in time. Or v_2 passes the zero-crossing line first if compared between two same phase points of v_1 and v_2 .
- Therefore, we say that v_2 leads v_1 by ϕ or that v_1 lags v_2 by ϕ .
- If $\phi \neq 0$, v_1 and v_2 are out of phase.
- If $\phi = 0$, v_1 and v_2 are in phase.



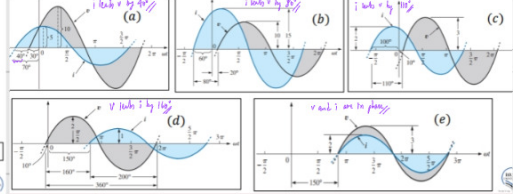
Problem 10

- Determine for each of the plots, which one is leading/lagging.



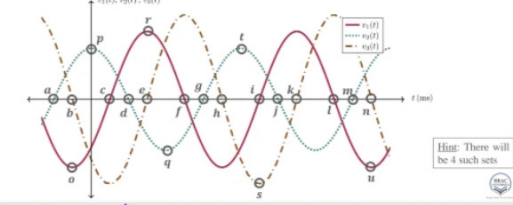
Problem 11

- Determine for each of the plots, which one (v or i) is leading/lagging and by how much.



Problem 12

- In the following figure, certain locations on the plots are marked with circles and labelled from a to u. Create sets by grouping the labels that correspond to the same phase.



- green wave: a, f, m, p, t, j
- yellow wave: b, h, n, e, k, i
- red wave: c, u, q, d, l

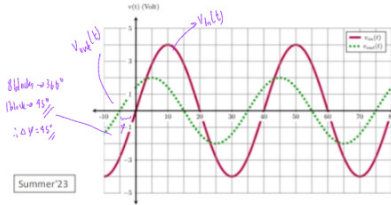
Q) Which one is leading and which one is lagging?

$V_s = 45 \cos(5\pi t + 36^\circ) = 45 \sin(5\pi t + 126^\circ)$
 $I_s = 80 \sin(5\pi t - 20^\circ)$
 $\# V_s$ leads I_s by $(126 - (-20)) = 146^\circ$
 $\# I_s$ lags V_s by 146°
 $\# I_s$ leads V_s by $(-20 - 126) = -146^\circ$
 → correct but not the convention

$-180^\circ \leq \text{phase}(\phi) \leq 180^\circ$
 phase difference $(\phi_A - \phi_B) \leq 180^\circ$
 A leads B by $(\phi_A - \phi_B)$
 $> 0, \leq 180$

Problem 12

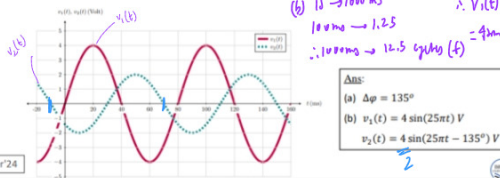
- Two ac voltages $v_1(t)$ and $v_2(t)$ from an ac circuit are plotted below. Which one is leading and by how much in degrees?



Problem 13

Two ac voltages $v_1(t)$ and $v_2(t)$ from an ac circuit are plotted below.

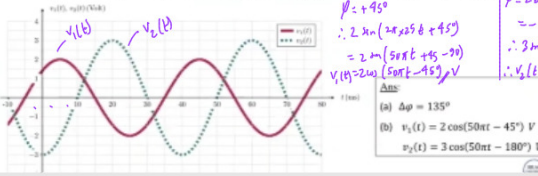
- a) Which one is leading and by how much in degrees?
b) Write analytical expressions for $v_1(t)$ and $v_2(t)$.



Problem 17

Two ac voltages $v_1(t)$ and $v_2(t)$ from an ac circuit are plotted below.

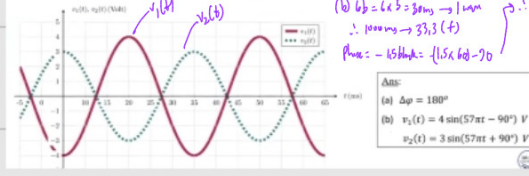
- a) Which one is leading and by how much in degrees?
b) Write analytical expressions for $v_1(t)$ and $v_2(t)$ in the form $A_m \cos(\omega t \pm \phi)$.



Problem 18

Two ac voltages $v_1(t)$ and $v_2(t)$ from an ac circuit are plotted below.

- a) Which one is leading and by how much in degrees?
b) Write analytical expressions for $v_1(t)$ and $v_2(t)$ in the form $A_m \sin(\omega t \pm \phi)$.

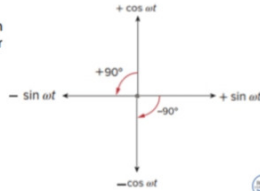


Sine-Cosine Conversion

- A sinusoid can be expressed in either sine or cosine form. When comparing two sinusoids, it is expedient to express both as either sine or cosine with positive amplitudes.

- The following trigonometric identities can be used to convert from sine to cosine or vice versa.

$$\begin{aligned}\sin(\omega t \pm 180^\circ) &= -\sin \omega t \\ \cos(\omega t \pm 180^\circ) &= -\cos \omega t \\ \sin(\omega t \pm 90^\circ) &= \pm \cos \omega t \\ \cos(\omega t \pm 90^\circ) &= \mp \sin \omega t\end{aligned}$$



Problem 14

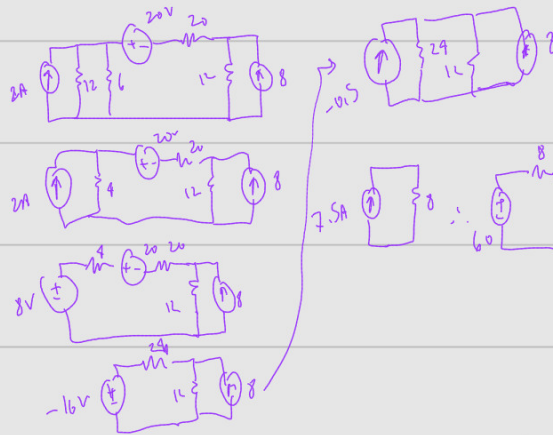
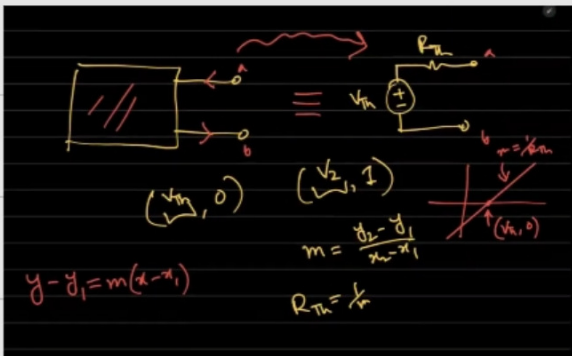
- For the following pairs of sinusoids, determine which one leads and by how much ($0 \leq \phi \leq 180^\circ$).

- $v_1(t) = 10 \cos(4t - 60^\circ)$ and $i(t) = 4 \sin(4t + 50^\circ)$
- $v_1(t) = 4 \cos(377t + 10^\circ)$ and $v_2(t) = -20 \cos 377t$
- $v_1(t) = 45 \sin(\omega t + 30^\circ)$ V and $v_2(t) = 50 \cos(\omega t - 30^\circ)$
- $i_1(t) = -4 \sin(377t + 55^\circ)$ and $i_2(t) = 5 \cos(377t - 65^\circ)$
- $x(t) = (13 \cos 2t + 5 \sin 2t)$ and $y(t) = 15 \cos(2t - 11.8^\circ)$

Hint: convert both the sinusoids into sine or cosine form $\rightarrow$ convert them into phasors $\rightarrow$ add them in frequency domain $\rightarrow$ convert them back in the time domain and compare

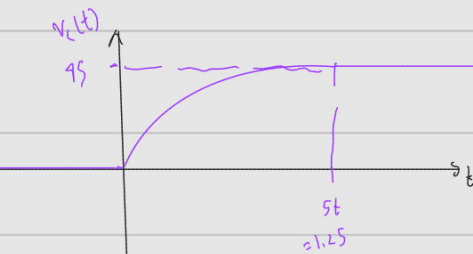
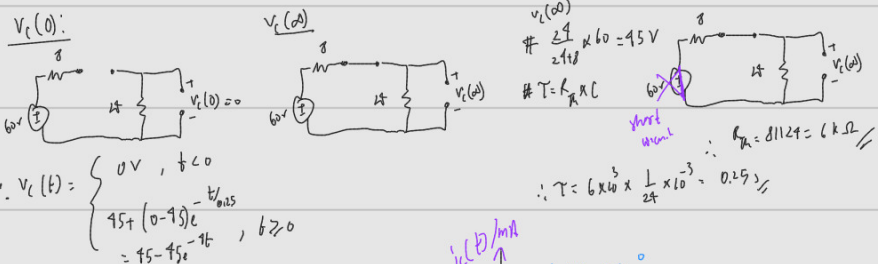
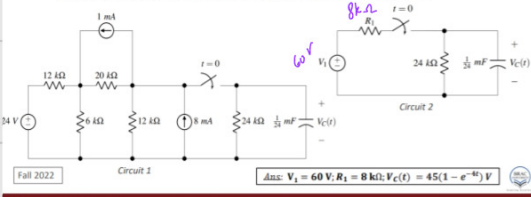
- Ans: I. i leads v by 20°
II. v_2 leads v_1 by 170°
III. v_2 leads v_1 by 30°
IV. i_2 leads i_1 by 150°
V. y leads x by 9.24°

I-V Characteristics

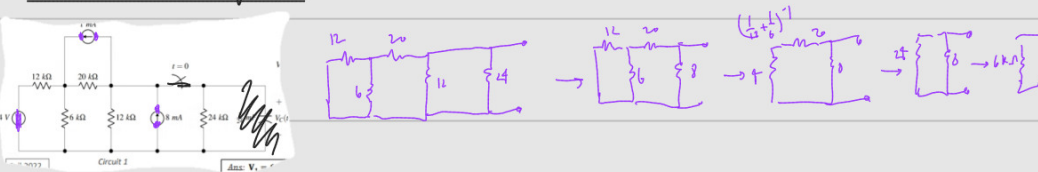


Problem 13

- Simplify the Circuit 1 below so that it takes the form of the Circuit 2. Determine the values of V_1 and R_1 . Perform transient analysis to determine $V_C(t)$ across the capacitor for $t > 0$.



Procm-2 of finding Rm

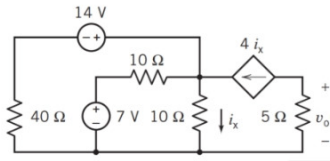


Superposition Theorem

Do not KVL dependent source

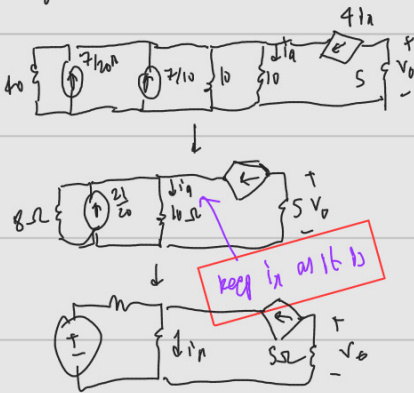
Problem 7

Use Superposition Principle to solve for v_o .

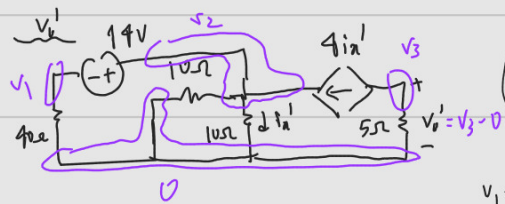


Ans: $v_o = 8 + 4 = 12V$

Using Source Transformation



By superposition, $v_o = v_o' + v_o''$, $i_x = i_x' + i_x''$



$$v_2 - v_1 = 14V$$

$$i_x' = v_2 / 10$$

$$v_o' = v_3$$

$$v_1 = v_2 - 14$$

$$\# v_3 \left(\frac{1}{5} \right) - 0 + 4i_x' = 0$$

$$v_3 \left(\frac{1}{5} \right) + 4 \left(\frac{v_2}{10} \right) = 0$$

$$\# (v_1 + v_2)$$

$$v_1 \left(\frac{1}{40} \right) + v_2 \left(\frac{1}{10} + \frac{1}{10} \right) - 4i_x' = 0$$

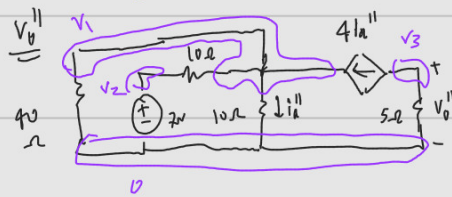
$$(v_2 - 14) \left(\frac{1}{40} \right) + v_2 \left(\frac{1}{10} + \frac{1}{10} \right) - 4 \left(\frac{v_2}{10} \right) = 0$$

$$v_2 \left(\frac{1}{40} + \frac{1}{10} + \frac{1}{10} - \frac{4}{10} \right) = \frac{14}{40}$$

$$v_2 = -2V$$

$$\therefore v_3 = \frac{16}{5} = 3.2V$$

$$\therefore v_o' = 4V$$



$$\# v_2 - 0 = 7V \quad \# v_3 \left(\frac{1}{5} \right) + 4i_x'' = 0$$

$$v_2 = 7$$

$$\# v_o'' = v_3$$

$$i_x'' = \frac{v_3}{10}$$

$$\# v_1 \left(\frac{1}{40} + \frac{1}{10} + \frac{1}{10} \right) - \frac{v_2}{10} - 4i_x'' = 0$$

$$v_1 \left(\frac{1}{40} + \frac{1}{10} + \frac{1}{10} \right) - \frac{7}{10} - 4 \left(\frac{v_3}{10} \right) = 0$$

$$\therefore v_1 \left(\frac{-4}{90} \right) = \frac{4}{10} \quad \therefore v_1 = -4V$$

$$\therefore v_3 \left(\frac{1}{5} \right) + 4 \left(\frac{-4}{10} \right) = 0$$

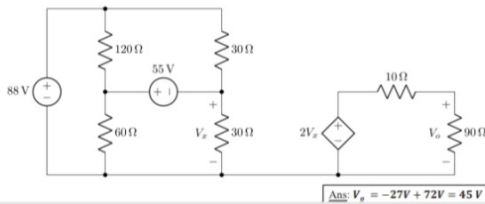
$$v_3 \left(\frac{1}{5} \right) = \frac{16}{10}$$

$$v_3 = 8V = v_o''$$

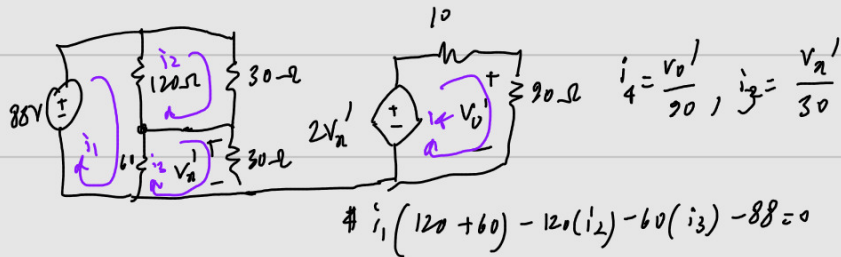
$$\therefore v_o = v_o' + v_o'' = 4 + 8 = 12V //$$

Problem 8

Use Superposition Principle to solve for v_o .



Ans: $v_o = -27V + 72V = 45V$



$$\# i_1(120 + 60) - 120(i_2) - 60(i_3) - 88 = 0$$

$$\# i_2(120 + 30) - 120(i_1) = 0$$

$$\# i_3(60 + 30) - 60(i_1) = 0$$

$$\# i_4(10 + 90) - 2V_x = 0$$

$$i_4(100) - 2(30i_3) = 0$$

$$i_4(100) - 2(30 \times \frac{4}{3}) = 0$$

$$i_4(100) - 80 = 0$$

$$i_4 = \frac{80}{100} = 0.8$$

$$\therefore v_o = v_o' + v_o'' = -27 + 72 = 45V //$$



$$i_4 = \frac{v_o''}{90}, \quad i_3 = \frac{v_o''}{30}$$

$$i_1(120 + 60) - 120(i_2) - 60(i_3) = 0$$

$$i_2(120 + 30) - 120(i_1) - 55 = 0$$

$$i_3(60 + 30) - 60(i_1) + 55 = 0$$

$$i_4(10 + 90) - 2V_x = 0$$

$$i_4(100) - 2(i_3 \times 30) = 0$$

$$i_1 = \frac{1}{6}, i_2 = \frac{1}{2}, i_3 = -\frac{1}{2}$$

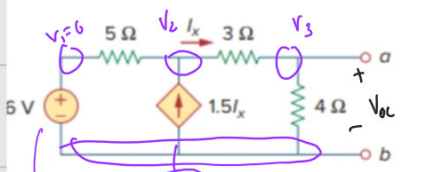
$$i_4(100) - 2(-\frac{1}{2} \times 30) = 0$$

$$i_4(100) + 30 = 0$$

$$i_4 = -0.3$$

$$\therefore v_o'' = -0.3 \times 90 = -27V //$$

$$\therefore v_o = v_o' + v_o'' = -27 + 72 = 45V //$$



grounded voltage source

Practice Problem 4.9

6V

5 Ω

3 Ω

4 Ω

1.5 I_x

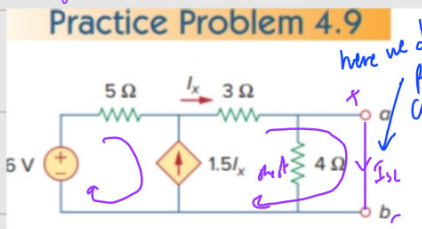
I_x

I_4

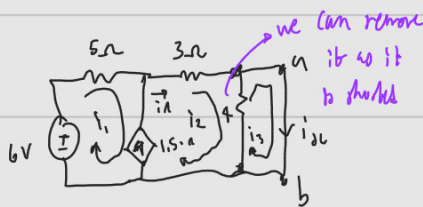
a

b

here we use KCL



here we do follow
parallel syn-
chronization



we can remove it as it is shaded

$$R_{Th} = \frac{5.33}{12} = 0.44 \Omega //$$

3 mesh, 1 supermesh (1,2)

$$\# \quad i_2 - i_1 = 1.5 i_2 \quad \# \quad i_{26} = i_3$$

$$\Rightarrow 1.5 i_2 \longrightarrow 0.5 i_2 + i_1 = 0$$

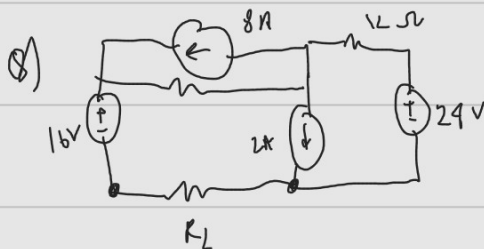
$$* \{i_1(5) - 6\} + \{i_2(3+4) - 4i_3\} = 0$$

$$\# i_3(4) - 4(i_2) = 0$$

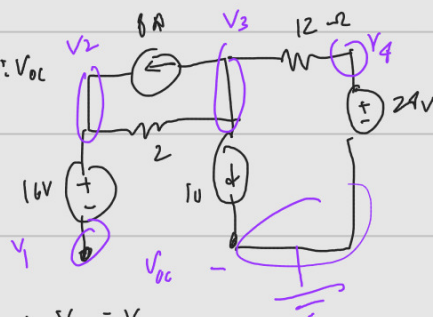
$$I_1 = -6A, I_2 = 12, I_3 = 12$$

We don't follow passive sign convention:

- ↳ During Source Transformation
- ↳ Test Current Source



Step-1 : V_{OC}



$$V_{DL} = V_1$$

$$\# \quad |b| = v_2 - v_1$$

$$A \begin{pmatrix} v_1(0) \\ v_2 \end{pmatrix} + \begin{pmatrix} 1 \\ -2 \end{pmatrix} - \frac{v_3}{2} - g = 0$$

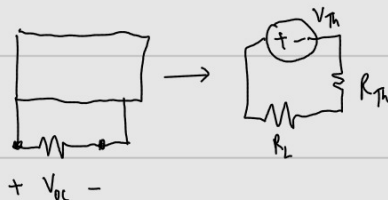
same eqⁿ but

$$A \quad \sqrt{3} \left(\frac{1}{12} + \frac{1}{2} \right) - \frac{\sqrt{4}^{24}}{2} - \frac{V_2}{2} + 10 = 0$$

$$V_4 = 24 \quad \checkmark$$

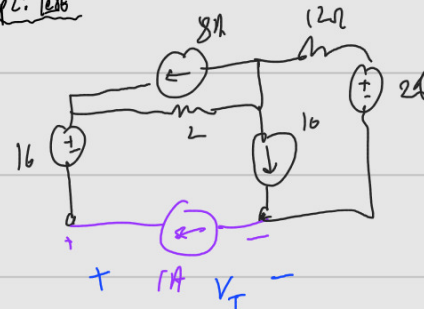
$$\therefore V_1 = 0V, V_2 = 16V, V_3 = 0$$

$\therefore V_{OL} = 0V \rightarrow$ hence put test current source //



4 nodes & 1 dependent (1,2)

Step 2: test



$$16 = V_2 - V_1$$

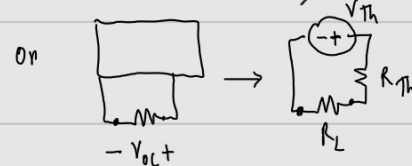
$$v_1(0) + v_2 \left(-\frac{1}{2} \right) - \frac{v_3}{2} - 8 = 0$$

$$V_3 \left(\frac{1}{12} + \frac{1}{2} \right) - \frac{V_4^{24}}{12} - \frac{V_2}{2} + 10 = 0$$

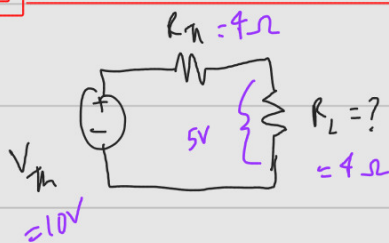
$$V_4 = 24$$

$$\therefore V_1 = V_T = 14V$$

$$R_{Th} = \frac{14}{1} = 14 \Omega$$



Maximum Power Transfer Theorem



It is when $R_L = R_{Th}$

$P_{L_{max}} = ?$ $P_{critic} = ?$

$$\text{eg } P_{\text{max}} = \frac{V_L^2}{R_1} = \frac{5^2}{4} = 6.25 \text{ W}$$

$$P_{\text{max (Load)}} = \frac{\left(\frac{V_{Th}}{2}\right)^2}{R_{Th}} = \frac{V_{Th}^2}{4R_{Th}}$$